The thermodynamics package*

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Abstract

A package, thermodynamics, is defined that makes typesetting quantities found in thermodynamics texts relatively simple. The commands are flexible and intended to be relatively intuitive. It handles several sets of notation for total, specific, and molar quantities; allows changes between symbols (e.g., A vs. F for Helmholtz free energy); and greatly simplifies the typesetting of symbols and partial derivatives commonly encountered in mixture thermodynamics. Changes of one's notes from one textbook to another can be achieved relatively easily by changing package options.

1 Introduction

The purpose of this package is to simplify the typesetting of equations in thermodynamics, specifically chemical engineering thermodynamics, which are often cumbersome to enter. For example, consider the following equation:

$$d\underline{U} = \left(\frac{\partial \underline{U}}{\partial \underline{S}}\right)_{\underline{V},\vec{n}} d\underline{S} + \left(\frac{\partial \underline{U}}{\partial \underline{V}}\right)_{\underline{S},\vec{n}} d\underline{V} + \sum_{i=1}^{C} \left(\frac{\partial \underline{U}}{\partial n_i}\right)_{\underline{S},\underline{V},n_{j\neq i}} dn_i. \tag{1}$$

This equation is pretty basic, and equations like it occur all the time in thermodynamics. Without this package, you might typeset it like this:

```
d\underline{U} =
  \left(\frac{\partial\underline{U}}
            {\partial\underline{S}}\right)_{\underline{V},\vec{n}}
  d\underline{S}
+ \left(\frac{\partial\underline{U}}
            {\partial\underline{V}}\right)_{\underline{S},\vec{n}}
  d\underline{V}
+ \sum_{i=1}^C \left(\frac{\partial\underline{U}}
     {\partial n_i}\right)_{\underline{S},\underline{V},n_{j\neq i}}
  dn_i.
```

This is a lot of code, and even then the output is slightly clunky:

$$d\underline{U} = \left(\frac{\partial \underline{U}}{\partial \underline{S}}\right)_{\underline{V},\vec{n}} d\underline{S} + \left(\frac{\partial \underline{U}}{\partial \underline{V}}\right)_{\underline{S},\vec{n}} d\underline{V} + \sum_{i=1}^{C} \left(\frac{\partial \underline{U}}{\partial n_i}\right)_{\underline{S},\underline{V},n_{j\neq i}} dn_i.$$

*This document corresponds to thermodynamics v1.00, dated 2022/04/29.

It is also frustratingly difficult to change one's notes or handouts from one textbook that uses, say, n_1 to denotes moles of component 1 to another textbook that uses N_1 for the same quantity, or perhaps denotes the total internal energy as U or U^t rather than $\underline{U}$. For example, if you wanted it to be

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,N_1,\ldots,N_n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,N_1,\ldots,N_n} dV + \sum_{i=1}^{n} \left(\frac{\partial U}{\partial N_i}\right)_{S,V,N_1,\ldots,[N_i],\ldots,N_n} dN_i$$

without changing any of your code—to update it across all handouts, exams, and homework sets after changing textbooks, say—you would be out of luck (or in for a lot of work).

With this package, you could reduce the code to typeset this equation to

```
\begin{equation}
   d\Ut = \Partial*{\Ut}{\St}{\Vt,\allNs} d\St
     + \Partial*{\Ut}{\Vt}{\St,\allNs} d\Vt
     + \sumall_i \Partial*{\Ut}{\Nt_i}{\St,\Vt,\allNsbut{i}} d\Nt_i
\end{equation}
```

and it will render similarly to Equation (1), including the shortened underscores and negative kerning. If you later decide to change the notation such that extensive properties are not underlined, you can do that without changing any of your code (just change a package option). Similarly, if you want $\vec{n}$ replaced by $n_1, \ldots, n_C$, you can do that with a package option, too.

The package handles second derivatives, too. For example,

```
\Partial*{\Hm}{T}{P} = T\Partial*{\Sm}{T}{P}
                     = -T\PartialSecond*{\Gm}{T}{P} = \cP
```

renders (using the default options)

$$\left(\frac{\partial H}{\partial T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_P = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_P = C_P$$

Similarly, macros are defined for mixed second partial derivatives that allow things like

```
\Partial*{\Gpm_i}{P}{T,\allNs}
  = \PartialMixSecond*{\Gt}{P}{\Nt_i}{T,\allNsbut{i}}
  = \PartialMixSecond*{\Gt}{\Nt_i}{P}{T,\allNsbut{i}}
  = \Partial*{Vt}{\Nt_i}{T,P,\allNsbut{i}} = \Vpm_i
```

which renders

$$\left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}} = \left(\frac{\partial^2 \underline{G}}{\partial P \partial n_i}\right)_{T,n_{j\neq i}} = \left(\frac{\partial^2 \underline{G}}{\partial n_i \partial P}\right)_{T,n_{j\neq i}} = \left(\frac{\partial \underline{V}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \overline{V}_i$$

using the defaults.

2 Using the Package

There are three categories of macros defined in this package: macros that produce symbols (or groups of them), macros that typeset derivatives, and macros that are used internally that the user need not know about. There are also several environments that allow the user to switch locally between different delimiters on partial derivatives.

2.1 Predefined Symbols

The macros used to produce symbols fall into five categories: extensive properties, molar properties, specific properties (i.e., per unit mass), partial molar properties, and shortcut macros (e.g., macros for the heat capacities, saturation pressure, and so forth). The macros corresponding to extensive, molar, and specific properties are shown in Table 1. Examples using the enthalpy are

\Ht
\Hm
\Hs
\Hpm

```
\[ \Ht \quad \Hm \quad \Hs \quad \Hpm_i. \]
```

Using the default package options, the above renders as

$$\underline{H} \quad H \quad \hat{H} \quad \overline{H}_i.$$

In addition, the properties in Table 2 are defined for convenience.

\Ut
\Um

How these symbols are rendered can be customized by package options. As long as the user consistently uses `\Ut` to render the total internal energy, `\Um` to render the molar internal energy, and so forth, switching notation from, say, $\underline{U}$ to U^t for extensive properties is trivial.

\cPt
\cVt
\cPs
\cVs
\cPpm
\cVpm

The heat capacities (see Table 2) are generally assumed to be molar (e.g., `\cP` is interpreted to be the molar heat capacity). To get the specific heat capacities, the macros `\cPs` and `\cVs` are provided, which by default render as $\hat{C}_P$ and $\hat{C}_V$, respectively. There are also extensive versions, so `\cPt` and `\cVt` will render as $\underline{C}_P$ and $\underline{C}_V$, respectively. Note that `\cP` and friends require you to surround `\text` with brackets if the portion with `\text` in it is not the last argument. For example, `\cP_i^\text{A}` will work as expected, but `\cP^\text{A}_i` will not: you need to use `\cP^{\text{A}}_i`. Using `\cP^\IG_i` will work as expected.

You can also get partial molar heat capacities via `\cPpm` and `\cVpm`, though the latter's mathematical definition is a bit hard to wrap one's head around:

```
\[ \cVpm_i = \Partial{\cVt}{\Nt_i}{T,P,\allNsbut{i}}
           = \frac{\partial}{\partial\Nt_i}
             \left[ -T\PartialSecond{\Ft}{T}{\Vt,\allNs}
             \right]_{T,P,\allNsbut{i}} \]
```

yields

$$\overline{C_{V,i}} = \left(\frac{\partial \underline{C}_V}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \frac{\partial}{\partial n_i}\left[-T\left(\frac{\partial^2 \underline{A}}{\partial T^2}\right)_{\underline{V},\vec{n}}\right]_{T,P,n_{j\neq i}}$$

2.2 Partial Molar Properties

\Upm
\Hpm

Partial molar quantities are typically defined with the suffix `pm`. For example, `\Upm` refers to the partial molar internal energy. There are two options for how to enter partial molar quantities: as commands or as super/subscripts. For example,

```
\[ \Upm{i} \quad \Upm[\IG]{i} \quad \Hpm_i \quad \Hpm^\IG_i \]
```

will typeset as

$$\overline{U}_i \quad \overline{U_i^{\mathrm{IG}}} \quad \overline{H}_i \quad \overline{H_i^{\mathrm{IG}}}$$

There are also partial molar heat capacities available via `\cPpm` and `\cVpm`. **Important**: The `\text` command defined by the `amstext` package is usually robust enough

Table 1. Commands defined in this package to represent extensive thermodynamic quantities and their molar and specific analogs. These macros should be used even if the symbol the user wishes to use does not match the command used (e.g., `\Ft` for total Helmholtz free energy even if it ends up being set as $\underline{A}$).

Property	Total	Molar	Specific	Partial Molar
Heat	`\Qt`	`\Qm`	`\Qs`	N/A
Work	`\Wt`	`\Wm`	`\Ws`	N/A
Total energy	`\Et`	`\Em`	`\Es`	`\Epm`
Internal energy	`\Ut`	`\Um`	`\Us`	`\Upm`
Enthalpy	`\Ht`	`\Hm`	`\Hs`	`\Hpm`
Entropy	`\St`	`\Sm`	`\Ss`	`\Spm`
Volume	`\Vt`	`\Vm`	`\Vs`	`\Vpm`
Helmholtz free energy	`\Ft`	`\Fm`	`\Fs`	`\Fpm`
Gibbs free energy	`\Gt`	`\Gm`	`\Gs`	`\Gpm`
Surface area	`\At`	`\Am`	`\As`	`\Apm`
Grand potential[a]	`\Lt`	`\Lm`	`\Ls`	`\Lpm`
Moles	`\Nt`	N/A	N/A	N/A
B (generic property)	`\Bt`	`\Bm`	`\Bs`	`\Bpm`
M (generic property)	`\Mt`	`\Mm`	`\Ms`	`\Mpm`

[a]The grand potential, $\underline{\Omega}(T, \underline{V}, \vec{\mu}) = \underline{U} - T\underline{S} - \sum_i^C \mu_i n_i$, is also called the Landau free energy by some authors.

that something like `\Um^\text{L}` will work as expected, without additional braces. This does *not* work for partial molar properties; for example, `\Hpm^\text{L}_i` will produce an error, as will `\Hpm_i^\text{L}`. The expression `\Hpm_i^{\text{L}}` will work as expected.

`\partialmolar` New partial molar properties can be defined for any "simple" symbol using the `\partialmolar` macro. "Simple" means it has no subscripts or superscripts. For `\Gpm` example, the macro for the partial molar Gibbs free energy is defined via the macro

```
\newcommand*{\Gpm}{\partialmolar{\Gibbs@symbol}}
```

A list of pre-defined macros for total, molar, specific, and partial molar quantities commonly used in thermodynamics is included in Table 1.

2.3 Other Predefined Symbols and Modifiers

There are a number of predefined symbols and modifiers. While these symbols could be defined or used without these macros, such use is not recommended: changing package options will result in inconsistencies if these macros are not used.

2.3.1 Heat Capacities, Compressibilities, and Expansivities

`\cP` The isobaric and isochoric heat capacities are produced with `\cP` and `\cV`, respec-
`\cV` tively. Four other measurable quantities are defined: the isothermal and isentropic
`\kappaS` compressibilities, `\kappaT` and `\kappaS`, respectively; and the isobaric and isentropic
`\kappaT` volume expansivities, `\alphaP` and `\alphaS`, respectively. Some textbooks use β in-
`\alphaP` stead of α for the volume expansivity to differentiate it from the *linear* expansivity;
`\alphaS`

Table 2. Convenience macros and their default symbols. These are generally "smart": for example, `\cP_i` renders as $C_{P,i}$, as expected, and `\cP_i^\circ` renders as $C^\circ_{P,i}$, also as expected. You can also reverse it: `\cP^\circ_i` becomes $C^\circ_{P,i}$.

Name	Macro	Sym.	Definition	Base Symbol Macro
Isobaric heat capacity	`\cP`[a]	C_P	$T\left(\frac{\partial S}{\partial T}\right)_P$	`\heatcapacitysymbol`
Isochoric heat capacity	`\cV`[a]	C_V	$T\left(\frac{\partial S}{\partial T}\right)_V$	`\heatcapacitysymbol`
Isothermal compressibility	`\kappaT`	κ_T	$-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$	`\compressibilitysymbol`
Isentropic compressibility	`\kappaS`	κ_S	$-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_S$	`\compressibilitysymbol`
Isobaric expansivity	`\alphaP`	α_P	$\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$	`\expansivitysymbol`
Isentropic expansivity	`\alphaS`	α_S	$\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_S$	`\expansivitysymbol`
Pure fugacity	`\fpure`	f	ϕP	
Mixture fugacity	`\fmix`	$\hat{f}$	$\hat{f}_i = x_i\hat{\phi}_i P$	
Saturation fugacity	`\fsat`	f^{sat}	$\phi^{\mathrm{sat}} P^{\mathrm{sat}}$	`\sat`
Pure fugacity coeff.	`\phipure`	ϕ	$\phi_i = \exp\left(\frac{1}{RT}\int_0^P V_i(T,p) - \frac{RT}{p}\,dp\right)$	
Mixture fugacity coeff.	`\phimix`	$\hat{\phi}$	$\hat{\phi}_i = \exp\left(\frac{1}{RT}\int_0^P \overline{V}_i(T,p,\vec{x}) - \frac{RT}{p}\,dp\right)$	
Henry's constant (rational)	`\Henryrat`	h	$\gamma_i^\infty f_i$	
Henry's constant (molal)	`\Henrymol`	$\mathcal{H}$	$M_s\gamma_i^\infty f_i$	
Rational activity coeff.	`\gammarat`	γ^*	γ/γ^∞	
Molal activity coeff.	`\gammamol`	$\gamma^\square$	$x_s\gamma/\gamma^\infty$	
Saturation fugacity coeff.	`\phisat`	ϕ^{sat}	$\phi(T, P^{\mathrm{sat}})$	`\sat`
Saturation pressure	`\Psat`	P^{sat}		`\sat`
Vapor pressure	`\Pvap`	Currently a synonym for `\Psat`[b]		
Standard state	`\std`[c]	$\circ$		
Standard pressure	`\Pstd`	P°		`\std`
Standard fugacity	`\fstd`	f°	$f(T, P^\circ)$	`\std`
Change on mixing	`\Deltamix`[d]	ΔM_{mix}	$M - \sum_i x_i M_i$	`\mixing`
Change on reaction	`\Deltarxn`[d]	ΔM_{rxn}	$\sum_i \nu_i M_i$	`\reaction`
Change on melting	`\Deltafus`[d]	ΔM^{fus}	$M^L - M^S$	`\fusion`
Change on boiling	`\Deltavap`[d]	ΔM^{vap}	$M^V - M^L$	`\vaporization`
Change on subliming	`\Deltasub`[d]	ΔM^{sub}	$M^V - M^S$	`\sublimation`

[a] Extensive and specific (per-unit-mass) versions are available as `\cPt` and `\cPs`, respectively, with similar macros for the isochoric heat capacity.

[b] If you want `\Pvap` to produce P^{vap} instead of P^{sat}, you should redefine the `\sat` macro.

[c] Typical usage would be `$\mu_i = \mu^\std_i + RT \log a_i$`, yielding $\mu_i = \mu_i^\circ + RT\log a_i$.

[d] The usual usage would be something like `$\Deltamix\Vm^\IGM = 0$`.

this can be changed by redefining \expansivitysymbol, which is done automatically by some of the package options that create notation specific to a particular textbook.

2.3.2 Fugacities and Fugacity Coefficients

\fpure \fmix \phipure \phimix

Different textbooks use different modifications of the symbol f for fugacity, so it is recommended to use the macro \fpure to denote the pure-component fugacity and \fmix to denote the mixture fugacity. Similarly, the pure-component fugacity coefficient should be generated with \phipure, and that in the mixture should be \phimix.

For example, the following markup is an example of a common equation in mixture thermodynamics:

```
\[ \fmix_j = x_j \phimix_j P = x_j \gamma_j \fpure_j. \]
```

With the default package options, this produces

$$\hat{f_j} = x_j \hat{\phi}_j P = x_j \gamma_j f_j.$$

With the Thompson package option, however, the same markup produces

$$\hat{f_j} = x_j \hat{\phi}_j P = x_j \gamma_j f_j^{\bullet}.$$

Similarly, the Prausnitz package option causes it to generate

$$f_j = x_j \phi_j P = x_j \gamma_j f_{\text{pure},j},$$

and the Sandler option causes it to generate

$$\bar{f_j} = x_j \bar{\phi}_j P = x_j \gamma_j f_j.$$

2.3.3 Activity Coefficients and Henry's Constants

\gammarat \gammamol \Henryrat \Henrymol

The activity coefficient can be generated with \gamma, as usual. The Henry's Law activity coefficients should be produced with \gammarat (rational basis) and \gammamol (molal basis). There are also macros to generate the Henry's law constants for both the rational basis (\Henryrat) and the molal basis (\Henrymol). These are interrelated:

```
\[ \fmix_i = x_i \gamma_i \fpure_i = x_i \gammarat_i \Henryrat_i
           = C_i \gammamol_i \Henrymol_i \]
```

produces

$$\hat{f_i} = x_i \gamma_i f_i = x_i \gamma_i^* h_i = C_i \gamma_i^{\square} \mathcal{H}_i$$

using the default options.

2.3.4 Saturation Properties

\Psat \Pvap \fsat \phisat \sat

The saturation pressure is generated with \Psat. The macro \Pvap is an alias for \Psat. The fugacity and fugacity coefficient at saturation are accessed via \fsat and \phisat, respectively. Package options can be used to change some of these to match the notation of specific textbooks.

The \sat macro is used "behind the scenes" as part of \Psat, \fsat, and \phisat, which produce P^{sat}, f^{sat}, and ϕ^{sat}, respectively. If you wanted to redefine them to be P^{vap}, f^{vap}, and ϕ^{vap}, you could simply redefine \sat with \renewcommand*{\sat}{{\text{vap}}}. This is done automatically using package option Sandler.

2.3.5 Standard States

\std \Pstd \fstd

The symbol ∘ (\circ) is used by default for standard states. This is intended to be easy to change should the user want to replace P° with $P^\ominus$, say. This is accessed via the \std macro. The macro \Pstd is defined as P^\std for convenience to denote standard pressures, and \fstd is defined for standard fugacities so as to ease implementation across textbooks.

The usual usage would be something like this:

```
\[ \Deltarxn\Gm = \sumall_i \nu_i \mu_i = \sumall_i \left[ \nu_i \mu^\std_i
   + \nu_i RT \log\left(\frac{\fmix_i}{\fstd_i}\right) \right]
   = \Deltarxn\Gm^\std + RT \log\left[\prodall_i a_i^{\nu_i} \right] \]
```

which produces

$$\Delta G_{\text{rxn}} = \sum_{i=1}^{C} \nu_i \mu_i = \sum_{i=1}^{C} \left[\nu_i \mu_i^\circ + \nu_i RT \log\left(\frac{\hat{f}_i}{f_i^\circ}\right)\right] = \Delta G_{\text{rxn}}^\circ + RT \log\left[\prod_{i=1}^{C} a_i^{\nu_i}\right]$$

with the default settings.

2.3.6 Changes on Mixing, Reaction, Fusion, Vaporization, and Sublimation

\Deltamix

Mixing properties are handled via the \Deltamix macro, and are used as in the following example:

```
\[ \Deltamix\Gm = \Gm - \sumall_i x_i \Gm_i = \Deltamix\Hm - T\Deltamix\Sm \]
```

which yields

$$\Delta G_{\text{mix}} = G - \sum_{i=1}^{C} x_i G_i = \Delta H_{\text{mix}} - T\Delta S_{\text{mix}}.$$

\mixing

Some textbooks (Sandler, Thompson) choose to typeset these with the word "mix" *before* the symbol, which is handled automatically by this package. The macro \mixing determines how the change in mixing label is rendered; the default is {\text{mix}}.

\Deltafus \Deltasub \Deltavap

The commands \Deltafus, \Deltasub, and \Deltavap typeset changes due to fusion (melting), sublimation (subliming), and vaporization (boiling), respectively. Their use is straightforward, viz.,

```
\[ \Deltasub\Hm = \Hm^V - \Hm^S = \Hm^V - \Hm^L + (\Hm^L - \Hm^S)
                = \Deltafus\Hm + \Deltavap\Hm \]
```

yielding

$$\Delta H^{\text{sub}} = H^V - H^S = H^V - H^L + (H^L - H^S) = \Delta H^{\text{vap}} + \Delta H^{\text{fus}}$$

with the default options. Note that some textbooks (e.g., Sandler) typeset these quantities quite differently.

\Deltaf \Deltarxn

The macros \Deltaf and \Deltarxn are intended to typeset the enthalpy or free energy of formation and reaction, respectively. For example,

```
\[ \Deltarxn\Hm^\std = \sumall_i \nu_i \Deltaf\Hm_i^\std \]
```

results in

$$\Delta H^{\circ}_{\text{rxn}} = \sum_{i=1}^{C} \nu_i \Delta H^{\circ}_{f,i}.$$

It is not anticipated that this command will be combined with something like a heat capacity, which already has a (potentially double) subscript, but as there is no "formation" heat capacity, that should not present a problem.

2.4 Residual and Excess Properties

\UR
\URt
\URs
\URpm
\UE
\UEt
\UEs
\UEpm
\excess
\residual

Additional macros are defined that make it easy to typeset the residual (also called "departure") and excess total, molar, specific, and partial molar properties. These macros follow the same pattern: `\UR`, `\URt`, `\URs`, and `\URpm` typeset the molar, total, specific, and partial molar residual internal energies, respectively, and by default expand to U^R, $\underline{U}^R$, $\hat{U}^R$, and $\overline{U}_i^R$ (the last is called as `\URpm{i}` or `\URpm_i`). Similarly, `\UE`, `\UEt`, `\UEs`, and `\UEpm` typeset the corresponding excess properties. The first character of the macros for other properties follow the same pattern as in Table 1.

The R and E characters are generated by the macros `\residual` and `\excess`, respectively. These macros can be redefined; for example, if you want `\SE`, which normally produces S^E, to give you S^{EX}—and let's be honest, who doesn't want that?[1]—then you can redefine it with

```
\renewcommand*{\excess}{{EX}}
```

or possibly

```
\renewcommand*{\excess}{{\mathrm{EX}}}
```

or even

```
\renewcommand*{\excess}{\text{EX}}
```

which cause `\SE` to expand to S^{EX}, S^{EX}, and S^{EX}, respectively.

It is generally not possible to use superscripts with the excess or residual properties; in the event the user needs this, the `\excess` and `\residual` macros can be used directly, viz.,

```
\begin{gather*}
  \Hm^{\excess,\std} = \HE(T,\Pstd) \\
\renewcommand*{\excess}{EX}
  \Hm^{\excess,\std} = \HE(T,\Pstd)
\end{gather*}
```

which yields

$$H^{E,\circ} = H^E(T, P^\circ)$$
$$H^{EX,\circ} = H^{EX}(T, P^\circ)$$

using the default options.

[1] You knew that joke was coming.

2.5 Partial Derivatives

\Partial
\Partial*

Partial derivatives are easily rendered using the \Partial command. There is a starred form (\Partial*) that additionally adjusts the spacing after the closing symbol to remove some of the space, anticipating that the following binary operator will overhang the subscripts. Compare the following:

```
\begin{gather*}
  \Partial{\Hm}{T}{P} = \cP \qquad \Partial*{\Hm}{T}{P} = \cP
\end{gather*}
```

which yields

$$\left(\frac{\partial H}{\partial T}\right)_P = C_P \qquad \left(\frac{\partial H}{\partial T}\right)_P = C_P$$

\PartialSecond
\PartialSecond*
\PartialMixSecond
\PartialMixSecond*

Second partial derivatives and mixed-second partial derivatives are typeset with the commands \PartialSecond and \PartialMixSecond, respectively. Like the first-order variety, these also have starred versions that remove the space immediately following the closing symbols, anticipating that the equals sign or other binary operator following the derivative will overhang the elements held constant. For example,

```
\[ \Vpm_i = \Partial*{\Vt}{\Nt_i}{T,P,\allNsbut{i}}
          = \PartialMixSecond*{\Gt}{\Nt_i}{P}{T,\allNsbut{i}}
          = \PartialMixSecond*{\Gt}{P}{\Nt_i}{T,\allNsbut{i}}
          = \Partial{\Gpm_i}{P}{T,\allNs} \]
```

looks like

$$\overline{V}_i = \left(\frac{\partial \underline{V}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial^2 \underline{G}}{\partial n_i \partial P}\right)_{T,n_{j\neq i}} = \left(\frac{\partial^2 \underline{G}}{\partial P \partial n_i}\right)_{T,n_{j\neq i}} = \left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}}.$$

\PartialBigg
\PartialSecondBigg
\PartialMixSecondBigg
\Partialbigg
\PartialSecondbigg
\PartialMixSecondbigg

There are instances (such as the line above) when \Partial causes parentheses that are slightly too tall but do not need to be—particularly when partial molar properties, specific quantities, or fugacities are involved. The macro \PartialBigg uses amsmath's \Biggl and \Biggr macros in place of \left and \right to size the parentheses accordingly; \Partiallbigg uses \biggl and \biggr in a similar fashion. For example, compare the following:

```
\[ \Vpm_i = \Partial*{\Vt}{\Nt_i}{T,P,\allNsbut{i}}
          = \Partial*{\Gpm_i}{P}{T,\allNs}
          = \PartialBigg*{\Gpm_i}{P}{T,\allNs}
          = RT\Partial*{\log\fmix_i}{P}{T,\allNs}
          = RT\,\PartialBigg*{\log\fmix_i}{P}{T,\allNs}
          = RT\,\Partialbigg{\log\fmix_i}{P}{T,\allNs} \]
```

which typesets as

$$\overline{V}_i = \left(\frac{\partial \underline{V}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}} = \left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}} = RT\left(\frac{\partial \log \hat{f}_i}{\partial P}\right)_{T,\vec{n}} = RT\left(\frac{\partial \log \hat{f}_i}{\partial P}\right)_{T,\vec{n}} = RT\left(\frac{\partial \log \hat{f}_i}{\partial P}\right)_{T,\vec{n}}$$

Note that a similar effect—possibly with other side effects—can be achieved with amsmath's \smash command, which has the effect of removing all vertical space associated with a particular character. Observe:

```
\[ \Vpm_i = \Partial{\Vt}{\Nt_i}{T,P,\allNsbut{i}}
          = \Partial{\smash{\Gpm_i}}{P}{T,\allNs} \]
```

produces

$$\overline{V}_i = \left(\frac{\partial \underline{V}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}}.$$

Note that the vertical spacing is not quite as good here as it was above. This can be fixed by using the optional argument to `\smash`, viz.,

```
\[ \Vpm_i = \Partial{\Vt}{\Nt_i}{T,P,\allNsbut{i}}
          = \Partial{\smash[t]{\Gpm_i}}{P}{T,\allNs} \]
```

$$\overline{V}_i = \left(\frac{\partial \underline{V}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial \overline{G}_i}{\partial P}\right)_{T,\vec{n}}.$$

It is possible to "fake" higher-order derivatives via some trickery. For example,

```
\[ \cPpm_i = T\Partial{\Spm_i}{T}{P,\allXs}
           = T\PartialSecond{\St}{T}{\Nt_i}{P,\allNsbut{i}}
           = -T\Partial{^3\Gt}{T^2\partial\Nt_i}{P,\allNsbut{i}} \]
```

gives

$$\overline{C_{P,i}} = T\left(\frac{\partial \overline{S}_i}{\partial T}\right)_{P,\vec{x}} = T\left(\frac{\partial^2 \underline{S}}{\partial T \partial n_i}\right)_{P,n_{j\neq i}} = -T\left(\frac{\partial^3 \underline{G}}{\partial T^2 \partial n_i}\right)_{P,n_{j\neq i}},$$

which is probably pretty close to what you wanted. Using this trickery with the package option `nosubscripts` will not work as well, and the use of third- and higher-order derivatives with this package should generally be considered unsupported.

2.6 Holding Constant the Number of Moles of Several Species

`\allNs` `\allNsbut` `\allmus` `\allmusbut` `\allXs` `\allXsbut` `\allYs` `\allYsbut` `\allMs` `\allMsbut` `\allWs` `\allWsbut`

It is common in thermodynamics to use notation such as

$$\overline{V}_k = \left(\frac{\partial \underline{V}}{\partial n_k}\right)_{T,P,n_{j\neq k}}$$

or perhaps

$$\overline{V}_k = \left(\frac{\partial \underline{V}}{\partial n_k}\right)_{T,P,n_1,\ldots,[n_k],\ldots,n_C}$$

to mean partial derivatives that hold the number of moles of each species constant *except* the one being changed. Similarly, a property determined with all mole fractions held constant might be written

$$C_P = \left(\frac{\partial H}{\partial T}\right)_{P,\vec{x}}$$

or perhaps

$$C_P = \left(\frac{\partial H}{\partial T}\right)_{P,x_1,\ldots,x_C}.$$

There are several macros that standardize such constructs. The `\allNs` macro expands to something meaning the number of moles of all species; by default, this is $\vec{n}$ (package option `moles-index`), but can be changed to $n_1, \ldots, n_C$ using the package option `moles-range`. Similarly, the macros `\allmus` and `\allmusbut` do the same but with n replaced by μ, and `\allMsbut` is the same with m instead of n. There are analogous macros for mole fractions, namely `\allXs` and `\allXsbut` for x and `\allYs` and `\allYsbut` for y, as well as `\allWs` and `\allWsbut` for mass fractions—these implicitly assume that all mole or mass fractions *except* the last are used as variables. The macros `\allNsbut`, `\allMsbut`, `\allmusbut`, `\allXsbut`, `\allYsbut`, and `\allWsbut` take an optional argument; for example,

```
\Partial{\Ht}{\Nt_1}{T,P,\allNsbut{1}} =
\Partial{\Ht}{\Nt_1}{T,P,\allNsbut[m]{1}} = \Hpm_1
```

typesets as

$$\left(\frac{\partial \underline{H}}{\partial n_1}\right)_{T,P,n_{j\neq 1}} = \left(\frac{\partial \underline{H}}{\partial n_1}\right)_{T,P,n_{m\neq 1}} = \overline{H_1}$$

Similarly,

```
\Partial{\Hm}{x_i}{T,P,\allXsbut{i}} =
\Partial{\Hm}{x_i}{T,P,\allXsbut[m]{i}} = \Hpm_i - \Hpm_\ncomponents
```

becomes

$$\left(\frac{\partial H}{\partial x_i}\right)_{T,P,x_{j\neq i,C}} = \left(\frac{\partial H}{\partial x_i}\right)_{T,P,x_{m\neq i,C}} = \overline{H_i} - \overline{H_C}$$

The user must supply their own redefinition if they wish to hold something other than `\ncomponents` constant in addition to the argument. Using the `moles-range` package option, for which `\allXsbut{k}` expands to $x_1, \ldots, [x_k], \ldots, x_{C-1}$ rather than $x_{j\neq k,C}$, the optional argument is ignored.

`\allbut` `\allbutlastand`

Users can define new "all but" macros using the `\allbut` and `\allbutlastand` commands. For example,

```
\newcommand*{\allNsbut}[2][j]{\allbut[#1]{#2}{\Nt}}
\newcommand*{\allXsbut}[2][j]{\allbutlastand[#1]{#2}{x}}
```

are the definitions of `\allNsbut` and `\allXsbut`, respectively.

2.7 Sums and Products

`\sumall` `\sumallbutlast` `\prodall`

It is common to require sums and products such as

$$\sum_{i=1}^{C} x_i = 1 \quad \text{or} \quad x_C = 1 - \sum_{i=1}^{C-1} x_i \quad \text{or} \quad \underline{G} = \sum_{j=1}^{C} \mu_j n_j \quad \text{and} \quad K = \prod_{k=1}^{C} a_k^{\nu_k}.$$

This package defines shortcuts to typeset such terms thus:

```
\[ \sumall_i x_i = 1 \quad \text{or} \quad
   x_\ncomponents = 1 - \sumallbutlast_i x_i \quad \text{or} \quad
   \Gt = \sumall_j \mu_j\Nt_j \quad \text{and} \quad
   K = \prodall_k a_k^{\nu_k}. \]
```

The symbol C can be changed by redefining `\ncomponents`. This is done automatically by some package options (e.g., `TesterModell` changes it to n; `Sandler` changes it to C; `Thompson` changes it to c).

Table 3. Options controlling which symbols to use by default. The macros `\Et`, `\Ut`, `\Ft`, `\Gt`, `\Ht`, `\At`, and `\Nt` represent the total energy, internal energy, Helmholtz free energy, Gibbs free energy, enthalpy, surface area, and number of moles, respectively. Symbols are shown as they would appear with the (default) option `intensive-plain`.

Option	\Et	\Ut	\Ft	\Gt	\Ht	\At	\Nt
EUAGHan	$\underline{E}$	$\underline{U}$	$\underline{A}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	n
EUAGHaN	$\underline{E}$	$\underline{U}$	$\underline{A}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	N
EUHAGan	(synonym for EUAGHan)						
EUHAGaN	(synonym for EUAGHaN)						
EUFGHAn	$\underline{E}$	$\underline{U}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	n
EUFGHAN	$\underline{E}$	$\underline{U}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	N
EEFGHAn	$\underline{E}$	$\underline{E}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	n
EEFGHAn	$\underline{E}$	$\underline{E}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	N
EEFGHan	$\underline{E}$	$\underline{E}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	n
EEFGHaN	$\underline{E}$	$\underline{E}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	N
EEAGHaN	$\underline{E}$	$\underline{E}$	$\underline{A}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	N
EUAGHAn	$\underline{E}$	$\underline{U}$	$\underline{A}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	n
EUAGHAN	$\underline{E}$	$\underline{U}$	$\underline{A}$	$\underline{G}$	$\underline{H}$	$\underline{A}$	N
EUFGHan	$\underline{E}$	$\underline{U}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	n
EUFGHaN	$\underline{E}$	$\underline{U}$	$\underline{F}$	$\underline{G}$	$\underline{H}$	$\underline{a}$	N

3 Loading the Package

To load the package with the defaults enabled, load it the usual way:

```
\usepackage{thermodynamics}
```

The package options loaded by default are `EUAGHan`, `subscripts`, `parentheses`, `intensive-plain`, and `moles-index`. These define, respectively, the default symbols to use for total energy, internal energy, Helmholtz free energy, and so forth; the manner of writing partial derivatives; the delimiters around partial derivatives; and the manner of denoting extensive, molar, and specific properties. The default behavior can be altered by options in the following section.

3.1 Package Options

There are three categories of options: options that affect which symbols are used, options that affect how symbols are decorated, and options that affect how partial derivatives are displayed. These are presented in turn.

3.1.1 Options that Change Symbol Sets

There are several options that choose the set of symbols to use for total energy, internal energy, Helmholtz free energy, and so forth. These are summarized in Table 3. The default is `EUAGHan`.

Using `EUAGHan` (the default), we might use the following markup:

```
\[ \Ft = \Ut - T\St = -P\Vt + \sum_i \mu_i \Nt_i + \sigma d\At
   \qquad \Hm = \Um + P\Vm \qquad \Et = \Ut + \frac12 mv^2 \]
```

which would look like

$$\underline{A} = \underline{U} - T\underline{S} = -P\underline{V} + \sum_i \mu_i n_i + \sigma d\underline{a} \qquad H = U + PV \qquad \underline{E} = \underline{U} + \frac{1}{2}mv^2.$$

Using the `EUFGHAN` option, the same markup would yield

$$\underline{F} = \underline{U} - T\underline{S} = -P\underline{V} + \sum_i \mu_i N_i + \sigma d\underline{A} \qquad H = U + PV \qquad \underline{E} = \underline{U} + \frac{1}{2}mv^2.$$

3.1.2 Options for Extensive vs. Molar Properties

There are four sets of notation that define how extensive properties are represented, as shown in Table 4. The default is `intensive-plain`, which (using the volume as an example) represents the total, molar, specific, and partial molar volumes, respectively, as $\underline{V}$, V, $\hat{V}$, and $\overline{V}_j$, respectively.

For example, the definition of the partial molar enthalpy would be different depending on which set of notation is used. The markup

```
\Hpm_i = \Partial{\Ht}{\Nt_i}{T,P,\allNsbut{i}}
       = \Partial{\Nt\Hm}{\Nt_i}{T,P,\allNsbut{i}}
```

yields the following, depending on the package option loaded:

$$\overline{H}_i = \left(\frac{\partial \underline{H}}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial nH}{\partial n_i}\right)_{T,P,n_{j\neq i}} \qquad \texttt{intensive-plain}$$

$$\overline{H}_i = \left(\frac{\partial H}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial n\underline{H}}{\partial n_i}\right)_{T,P,n_{j\neq i}} \qquad \texttt{extensive-plain}$$

$$\overline{H}_i = \left(\frac{\partial H^t}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial nH}{\partial n_i}\right)_{T,P,n_{j\neq i}} \qquad \texttt{extensive-superscript}$$

$$\overline{H}_i = \left(\frac{\partial H}{\partial n_i}\right)_{T,P,n_{j\neq i}} = \left(\frac{\partial nh}{\partial n_i}\right)_{T,P,n_{j\neq i}} \qquad \texttt{intensive-lowercase}$$

The use of `intensive-lowercase` is strongly discouraged.

Note that the number of moles can be changed from n to N via the options in the previous section.

3.1.3 Options Affecting Partial Derivatives

There are several options that change how partial derivatives are rendered. First are the options that affect the delimiters. We will use the following code as an example:

```
\[ \Partial*{\Vm}{T}{P} = \PartialMixSecond{\Gm}{T}{P}{}
                        = \PartialMixSecond{\Gm}{P}{T}{}
                        = -\Partial{\Sm}{P}{T}. \]
```

Using the `parentheses` option (the default), this gives

$$\left(\frac{\partial V}{\partial T}\right)_P = \left(\frac{\partial^2 G}{\partial T \partial P}\right) = \left(\frac{\partial^2 G}{\partial P \partial T}\right) = -\left(\frac{\partial S}{\partial P}\right)_T.$$

Table 4. Notation sets that can be set using the options `intensive-plain` (the default), `extensive-plain`, `extensive-superscript`, and `intensive-lowercase`, respectively. Note that specific quantities are generally assumed to look like the molar symbol with a caret on top.

Option	`\Vt`	`\Vm`	`\Vs`	`\Vpm_i`
`intensive-plain`	$\underline{V}$	V	$\hat{V}$	$\overline{V}_i$
`extensive-plain`	V	$\underline{V}$	$\hat{V}$	$\overline{V}_i$
`extensive-superscript`	V^t	V	$\hat{V}$	$\overline{V}_i$
`intensive-lowercase`	V	v	$\hat{v}$	$\overline{v}_i$

The option `brackets` changes the output to

$$\left[\frac{\partial V}{\partial T}\right]_P = \left[\frac{\partial^2 G}{\partial T \partial P}\right] = \left[\frac{\partial^2 G}{\partial P \partial T}\right] = -\left[\frac{\partial S}{\partial P}\right]_T .$$

The option `bar` changes the output to

$$\left.\frac{\partial V}{\partial T}\right|_P = \frac{\partial^2 G}{\partial T \partial P} = \frac{\partial^2 G}{\partial P \partial T} = -\left.\frac{\partial S}{\partial P}\right|_T .$$

The option `plain-derivatives` eliminates all delimiters; this forces the `nosubscripts` option. The output in this case is

$$\frac{\partial V(T,P)}{\partial T} = \frac{\partial^2 G(T,P)}{\partial T \partial P} = \frac{\partial^2 G(P,T)}{\partial P \partial T} = -\frac{\partial S(P,T)}{\partial P}$$

Accompanying the `plain-derivatives` option is the `nosubscripts` option, which overrides the default option `subscripts`. This option makes partial derivatives such as

$$\left(\frac{\partial V}{\partial P}\right)_T \qquad (\texttt{subscripts} \text{ option}),$$

and instead renders them

$$\left(\frac{\partial V(P,T)}{\partial P}\right) \qquad (\texttt{nosubscripts} \text{ option}).$$

Combined with `plain-derivatives`, this would give

$$\frac{\partial V(P,T)}{\partial P} \qquad (\texttt{nosubscripts} \text{ and } \texttt{plain-derivatives} \text{ options}).$$

The order of the variables is determined by the arguments given: it is always written with the variable that is changing first, and the other variables in the order given in the final argument to `\Partial` and friends. This will result in things like the following:

```
\[ \Partial{\Vm}{T}{P} = \PartialMixSecond{\Gm}{T}{P}{}
    = \PartialMixSecond{\Gm}{P}{T}{} = -\Partial{\Sm}{P}{T}, \]
```

which would produce (using `plain-derivatives`)

$$\frac{\partial V(T,P)}{\partial T} = \frac{\partial^2 G(T,P)}{\partial T \partial P} = \frac{\partial^2 G(P,T)}{\partial P \partial T} = -\frac{\partial S(P,T)}{\partial P}$$

This is not unclear, but the variables appear in a different order.

`thermoparentheses` `thermobrackets` `thermobar` `thermoplain` `thermosubscripts` `thermoNOsubscripts`

If you want to use parentheses *locally*, even though your overall document uses another delimiter, the `thermoparentheses` environment will do that. Similarly, `thermobrackets` will temporarily switch to brackets, `thermobar` will temporarily switch to a tailing vertical bar, and `thermoplain` will remove delimiters altogether.

The environments `thermosubscripts` and `thermoNOsubscripts` force the use or disuse of subscripts, respectively.

3.1.4 Options Regarding the Number of Moles

`\allNs` `\allNsbut`

The default option `moles-index` defines the macro `\allNs` to expand to $\vec{n}$ and the macro `\allNsbut{i}` to expand to $n_{j\neq i}$. You can change the dummy index: `\allNsbut[k]{i}` expands to $n_{k\neq i}$ by default. This is typically not necessary, however: if you type `\allNsbut{j}`, the package will figure out that you want $n_{k\neq j}$ rather than $n_{j\neq j}$. The time to use the optional argument is in situations such as

$$\left(\frac{\partial \mu_j}{\partial n_k}\right)_{n_{i\neq k}},$$

which is incorrect if the dummy index j is used in place of the i.

You can change these to expand to ranges using the `moles-range` option, which renders `\allNs` as $n_1, \ldots, n_C$ and `\allNsbut{i}` as $n_1, \ldots, [n_i], \ldots, n_C$. The optional argument is ignored in this set of notation. Examples of these options are shown in Table 5.

`\ncomponents`

You can change the symbol for the number of components (default: C) by redefining the macro `\ncomponents`.

3.1.5 Other Options

`\dbar`

The default for path-dependent one-forms (often called "inexact differentials") is `\dbar`, which looks like đ. This can be changed, if desired, to a delta (δ) with the `delta` option to the package.

It should be noted that the `\dbar` macro is not very stable: changing the typeface to something not supported will probably ruin it, as the kerning is very font-specific. This package currently supports Computer Modern, Times, Palatino, Bitstream Charter, Garamond, and Utopia, but other typefaces may require a manual redefinition.

3.1.6 Options for Specific Textbooks

There are several options that load package options and/or redefine particular commands to match the notation in a particular textbook. So far the following textbooks are supported:

Table 5. Illustration of the `moles-index` and `moles-range` options and their effects on `\allNs` and `\allNsbut`.

Macro[a]	`moles-index`	`moles-range`
`\allNs`	$\vec{n}$	$n_1,\ldots,n_C$
`\allNsbut{1}`	$n_{j\neq 1}$	$n_2,\ldots,n_C$
`\allNsbut{i}`	$n_{j\neq i}$	$n_1,\ldots,[n_i],\ldots,n_C$
`\allNsbut{j}`	$n_{k\neq j}$	$n_1,\ldots,[n_j],\ldots,n_C$
`\allNsbut{\ncomponents}`	$n_{j\neq C}$	$n_1,\ldots,n_{C-1}$
`\allNsbut[k]{i}`	$n_{k\neq i}$	$n_1,\ldots,[n_i],\ldots,n_C$
`\allXs`	$\vec{x}$	$x_1,\ldots,x_C$
`\allXsbut{1}`	$x_{j\neq 1,C}$	$x_2,\ldots,x_{C-1}$
`\allXsbut{i}`	$x_{j\neq i,C}$	$x_1,\ldots,[x_i],\ldots,x_{C-1}$
`\allXsbut{j}`	$x_{k\neq j,C}$	$x_1,\ldots,[x_j],\ldots,x_{C-1}$
`\allXsbut{\ncomponents-1}`	$x_{j\neq C-1,C}$	$x_1,\ldots,x_{C-2}$
`\allXsbut[k]{\ncomponents-1}`	$x_{k\neq C-1,C}$	$x_1,\ldots,x_{C-2}$
`\allXsbut{\ncomponents}`[b]	$x_{j\neq C}$	$x_1,\ldots,x_{C-1}$

[a] You may use C directly instead of `\ncomponents` here, but then it will not change $C-1$ to $C-2$ (or, say, $M-1$ to $M-2$) if you want to redefine `\ncomponents` later.

[b] This would typically be used to denote something like $\underline{G}(T,P,n,x_1,\ldots,x_{C-1})$ rather than in a subscript, but it looks silly if we don't handle this case this way.

Bejan Notation used by Bejan, *Advanced Engineering Thermodynamics*, Third Edition. Wiley: Hoboken, 2006. Loads non-default package options `EUFGHAN`, `intensive-lowercase`, and `delta`. Also swaps the definitions of `\@specific` and `\@intensive` and redefines `\cV`, `\cVs`, `\cVt`, `\expansivitysymbol`, and `\ncomponents` to match his notation.

CBK Notation used by Çengel, Boles, and Kanoğlu, *Thermodynamics: An Engineering Approach*, Ninth Edition. McGraw Hill: Singapore, 2020. Loads the non-default package options `EUAGHAN` and `intensive-lowercase`; also redefines partial molar, specific, and molar properties' notation to fit theirs, and redefines `\pressure@symbol`, `\Deltarxn`, `\compressibilitysymbol`, and `\expansivitysymbol` to fit their usage. Their prodigal symbols for specific and total volume are not supported.

ElliottLira Notation used by Elliott and Lira, *Introductory Chemical Engineering Thermodynamics*, Second Edition. Prentice Hall: Upper Saddle River, 2012. Loads the default package options and redefines `\allcomponents` and `\Deltarxn` to fit their notation.

Koretsky Notation used by Koretsky, *Engineering and Chemical Thermodynamics*, Second Edition, Wiley: New Caledonia, 2013. Loads the non-default package options `EUAGHAn`, `brackets`, `intensive-lowercase`, and `delta`; modifies the `intensive-lowercase` defaults to make uppercase partial molar properties to match Koretsky's notation and redefines `\expansivitysymbol`, `\IS`, `\residual`, `\IG`, `\IGM`, `\Deltafus`, `\Deltasub`, `\Deltavap`, `\Henryrat`, `\gammarat`, `\Deltafus`, `\Deltavap`, and `\Deltasub` to match his use. Also redefines the fugacity coefficients to use φ instead of ϕ.

MSBB Notation used by Moran, Shapiro, Boettner, and Bailey, *Fundamentals of Engineering Thermodynamics*, Eighth Edition. Wiley: Kendallville, 2014. Loads the non-default package options `EUFGHAn`, `intensive-lowercase`, and `delta`; removes the left parenthesis in partial derivatives and redefines `\IGM`, `\IG`, `\expansitivitysymbol`, `\allcomponents`, `\allbut`, `\Ft`, `\Helmholtz@symbol`, `\@intensive`, `\@specific`, `\fmix`, `\phimix`, and `\pressure@symbol` to fit their somewhat ill-advised notation. Also resets `\partialmolar` to match their use.

Prausnitz Notation used by Prausnitz, Lichtenthaler, and de Azevedo, *Molecular Thermodynamics of Fluid-Phase Equilibria*, Third Edition, Pearson, 1998. Loads the non-default package option `intensive-lowercase` and redefines `\fmix`, `\phimix`, `\fpure`, `\phipure`, `\residual`, `\allcomponents`, and `\allbut` to fit their notation.

Sandler Notation used by Sandler, *Chemical, Biochemical, and Engineering Thermodynamics*, Fifth Edition. Wiley: Hoboken, 2017. Loads non-default package options `EUAGHaN` and `extensive-plain`; also redefines `\sat`, `\excess`, `\residual`, `\ncomponents`, `\fmix`, `\fstd`, `\phimix`, `\allcomponents`, `\IS`, `\Deltamix`, `\Deltarxn`, `\Deltasub`, `\Deltafus`, `\Deltavap`, and `\Henryrat`. Also redefines `\cV`, `\cP`, `\cVt`, and `\cPt`.

SVNAS Notation used by Smith, Van Ness, Abbott, and Swihart, *Introduction to Chemical Engineering Thermodynamics*, Ninth Edition. McGraw-Hill: Boston, 2021. Loads the non-default package option `extensive-superscript` and redefines `\allcomponents`, `\allbut`, `\IG`, `\IGM`, `\IS`, `\expansitivitysymbol`, and `\Deltarxn` to fit their notation. The extensive heat capacities are also redefined, as they do not use such an entity.

TesterModell Notation used by Tester and Modell, *Thermodynamics and Its Applications*, Third Edition, Prentice Hall: Upper Saddle River, 1997. Loads the non-default package options `EUAGHaN` and `delta`; also redefines `\ncomponents` to be n and redefines `\allcomponents`, `\allbut`, and `\allbutlastand` to fit their (somewhat inconsistent) notation. Also redefines `\IG`, `\IGM`, `\IS`, `\excess`, `\reaction`, `\Henryrat`, `\Henrymol`, `\gammarat`, and `\gammamol` to fit their style.

Thompson Notation used by Thompson, *A Unified Introduction to Chemical Engineering Thermodynamics*, Stillwater Press: Orono, 2000. Loads the non-default package options `EUAGHAn` and `delta`; also redefines `\excess`, `\residual`, `\allcomponents`, `\ncomponents`, `\IS`, `\IG`, `\IGM`, `\fpure`, `\phipure`, `\mix`, `\Deltamix`, `\Deltarxn`, `\@intensive`, and `\allbut` to match his notation. Note that he uses c, n_C, and n for the number of components in various places in the book; I chose c for the definition of `\ncomponents`, but it is impossible to be completely consistent with his notation.

There may well be some inconsistencies between the notation in these books and the symbols used here. I will fix such inconsistencies as I become aware of them.

4 Implementation

This package requires the amstext package, as \text is used to handle \sat, \IS, \IG, \IGM, \Henrymol, \fusion, \reaction, \vaporization, and \sublimation by default, as well as several other macros defined by package options.

```
\RequirePackage{amstext}
```

4.1 Symbols Controlled by Package Options

We set some symbols prior to declaring the package options. The default symbols follow package option EUAGHan, even though the macros follow the option EUFGHAN.

\dbar The way \dbar is defined depends on the typeface you are using. We try to determine, at \begin{document}, which typeface you chose based on the packages that are loaded and some of their internal definitions. The thermodynamics package currently supports Computer Modern (the default or through lmodern), Palatino (through pxfonts or newpxmath), Times (through txfonts, mathptmx, or newtxmath), Utopia (through mathdesign), Bitstream Charter (through mathdesign), and Garamond (through mathdesign). Definitions of \dbar (with \newcommand* or \providecommand*) in the preamble will override the ones here.

```
\AtBeginDocument{
    \@ifpackageloaded{pxfonts}{%
        \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-10mu d}}%
    }{}
    \@ifpackageloaded{newpxmath}{%
        \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-10mu d}}%
    }{}
    \@ifpackageloaded{txfonts}{%
        \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-11mu d}}%
    }{}
    \@ifpackageloaded{mathptmx}{%
        \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-10mu d}}%
    }{}
    \@ifpackageloaded{newtxmath}{%
        \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-12mu d}}%
    }{}
    \@ifpackageloaded{mathdesign}{%
        \def\@charter{mdbch}%
        \def\@utopia{mdput}%
        \def\@garamond{mdugm}%
        \ifx\MD@default@family\@utopia
            \providecommand*{\dbar}{{\mkern8mu\mathchar'26\mkern-20mu d}}%
        \fi
        \ifx\MD@default@family\@charter
            \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-15mu d}}%
        \fi
        \ifx\MD@default@family\@garamond
            \providecommand*{\dbar}{{\mkern5mu\mathchar'26\mkern-17mu d}}%
        \fi
    }{}
    % Defaults to Computer Modern
    \providecommand*{\dbar}{{\mkern3mu\mathchar'26\mkern-12mu d}}
```

```
}
```

Symbols are defined for the total energy, internal energy, Helmholtz free energy, Gibbs free energy, enthalpy, entropy, surface area, volume, number of moles, heat, and work; these are E, U, A, G, H, S, a, V, n, Q, and W, respectively. These commands should not be used on their own, but rather accessed through the macros `\Ut`, `\Um`, and `\Us` (using the internal energy as an example).

`\totalenergy@symbol`
`\internalenergy@symbol`
`\Helmholtz@symbol`
`\Gibbs@symbol`
`\Landau@symbol`
`\enthalpy@symbol`
`\entropy@symbol`
`\area@symbol`
`\volume@symbol`
`\mole@symbol`
`\heat@symbol`
`\work@symbol`
`\temperature@symbol`
`\pressure@symbol`

The default symbols are not intended to be easy to change—the intended mechanism is through package options. If you want to use a non-standard symbol that is not available through one of the package options, you can redefine these macros inside `\makeatletter`...`\makeatother`. For example,

```
\makeatletter
   \renewcommand*{\Helmholtz@symbol}{H}
   \renewcommand*{\enthalpy@symbol}{h}
\makeatother
```

would define the ill-advised notation that I have nonetheless heard of that uses H for Helmholtz free energy and h for enthalpy. Note that the macros for temperature and pressure are only used inside the definitions of the compressibilities, expansivitites, and heat capacities; there is no user-level macro for the temperature or the pressure, so it is up to the user to use consistent symbols for those properties.

```
\newcommand*{\totalenergy@symbol}{E}
\newcommand*{\internalenergy@symbol}{U}
\newcommand*{\Helmholtz@symbol}{A}
\newcommand*{\Gibbs@symbol}{G}
\newcommand*{\Landau@symbol}{\Omega}
\newcommand*{\enthalpy@symbol}{H}
\newcommand*{\entropy@symbol}{S}
\newcommand*{\area@symbol}{a}
\newcommand*{\volume@symbol}{V}
\newcommand*{\mole@symbol}{n}
\newcommand*{\heat@symbol}{Q}
\newcommand*{\work@symbol}{W}
\newcommand*{\temperature@symbol}{T}
\newcommand*{\pressure@symbol}{P}
```

`\thermo@underline`
`\thermo@overline`

We then define two macros and several lengths that we shall use when drawing rules above or below a symbol. The default is to use underlined symbols for extensive quantities, plain symbols for molar quantities, and carets for specific quantities, but this can be changed using package options.

```
\newcommand*{\thermo@underline}[1]{%
  \mkern1mu\underline{\mkern-1mu #1\mkern-4mu}\mkern4mu%
}
\newcommand*{\thermo@overline}[1]{%
  \mkern2mu\overline{\mkern-2mu #1\mkern-1mu}\mkern1mu%
}
```

`\PartialOpen`
`\PartialClose`
`\PartialEmptyClose`

We define three commands to use to denote the beginning and end of partial derivatives. These symbols can be customized by package options. Default is `parentheses`,

meaning that `\[ \Partial{f}{x}{y} \]` renders as

$$\left(\frac{\partial f}{\partial x}\right)_y$$

with the defaults. The macro `\PartialEmptyClose` is used when the last argument to `\Partial` is empty, which is important for the `bar` option to the document class or inside the `thermobar` environment.

```
\newcommand*{\PartialOpen}{(}
\newcommand*{\PartialClose}{)}
\newcommand*{\PartialEmptyClose}{)}
```

4.2 Package Options

We declare a bunch of options for which sets of symbols to use. These are summarized in Table 3.

```
\DeclareOption{EUAGHan}{}% the default
\DeclareOption{EUAGHaN}{\renewcommand*{\mole@symbol}{N}}%
\DeclareOption{EUHAGan}{\ExecuteOptions{EUAGHan}}
\DeclareOption{EUHAGaN}{\ExecuteOptions{EUAGHaN}}
\DeclareOption{EUFGHAn}{% this is my favorite set of symbols
  \renewcommand*{\Helmholtz@symbol}{F}%
  \renewcommand*{\area@symbol}{A}%
}
\DeclareOption{EUFGHAN}{% this is my second favorite set of symbols
  \renewcommand*{\Helmholtz@symbol}{F}%
  \renewcommand*{\area@symbol}{A}%
  \renewcommand*{\mole@symbol}{N}%
}
\DeclareOption{EEFGHAn}{% this treats all energy the same way(?)
  \renewcommand*{\internalenergy@symbol}{E}%
  \renewcommand*{\Helmholtz@symbol}{F}%
  \renewcommand*{\area@symbol}{A}%
}
\DeclareOption{EEFGHAN}{% this treats all energy the same way(?)
  \renewcommand*{\internalenergy@symbol}{E}%
  \renewcommand*{\Helmholtz@symbol}{F}%
  \renewcommand*{\area@symbol}{A}%
  \renewcommand*{\mole@symbol}{N}%
}
\DeclareOption{EEFGHan}{%
  \renewcommand*{\internalenergy@symbol}{E}%
  \renewcommand*{\Helmholtz@symbol}{F}
}
\DeclareOption{EEFGHaN}{%
  \renewcommand*{\internalenergy@symbol}{E}%
  \renewcommand*{\Helmholtz@symbol}{F}
  \renewcommand*{\mole@symbol}{N}%
}
\DeclareOption{EEAGHan}{%
  \renewcommand*{\internalenergy@symbol}{E}%
}
\DeclareOption{EEAGHaN}{%
```

```
  \renewcommand*{\internalenergy@symbol}{E}%
  \renewcommand*{\mole@symbol}{N}%
}
\DeclareOption{EUAGHAn}{% Helmholtz free energy and area look the same here
  \renewcommand*{\area@symbol}{A}%
}
\DeclareOption{EUAGHAN}{% Helmholtz free energy and area look the same here
  \renewcommand*{\area@symbol}{A}%
  \renewcommand*{\mole@symbol}{N}%
}
\DeclareOption{EUFGHan}{%
  \renewcommand*{\Helmholtz@symbol}{F}%
}
\DeclareOption{EUFGHaN}{%
  \renewcommand*{\Helmholtz@symbol}{F}%
  \renewcommand*{\mole@symbol}{N}%
}
```

The `delta` option redefines `\dbar` to produce the symbol δ. The default is to use a d with a slash through it ($đ$) for inexact differentials unless the user overrides it with this option. The macro could also be redefined manually, of course.

```
\DeclareOption{delta}{\let\dbar\delta}
```

`\@extensive` `\@intensive` `\@specific`

Next, we define options for the set of notation. The default is `intensive-plain`, which produces things like V for molar volume, $\underline{V}$ for total volume, and $\hat{V}$ for specific volume. These are defined via the internal macros `\@extensive`, `\@intensive`, and `\@specific`.

```
\let\@extensive\relax
\let\@intensive\relax
\let\@specific\hat

\DeclareOption{extensive-plain}{%
    \let\@extensive\relax
    \let\@intensive\thermo@underline
}
\DeclareOption{intensive-plain}{% the default
    \let\@extensive\thermo@underline
    \let\@intensive\relax
}
\DeclareOption{intensive-lowercase}{% PLEASE don't use this!
    \let\@extensive\MakeUppercase
    \let\@intensive\MakeLowercase
    \def\@specific#1{\MakeLowercase{\hat #1}}
    \AtEndOfPackage{
        \renewcommand*{\partialmolar}[1]{%
            \gdef\pm@symbol{\MakeLowercase #1}\generic@pm}
        \renewcommand*{\heatcapacitysymbol}{c}
    }
}
\DeclareOption{extensive-superscript}{%
    \gdef\@extensive#1{{#1}^t}%
    \let\@intensive\relax%
    \AtEndOfPackage{%
      \renewcommand*{\URt}{\internalenergy@symbol^{\residual,t}}
```

```
      \renewcommand*{\HRt}{\enthalpy@symbol^{\residual,t}}
      \renewcommand*{\FRt}{\Helmholtz@symbol^{\residual,t}}
      \renewcommand*{\GRt}{\Gibbs@symbol^{\residual,t}}
      \renewcommand*{\VRt}{\volume@symbol^{\residual,t}}
      \renewcommand*{\SRt}{\entropy@symbol^{\residual,t}}
      \renewcommand*{\UEt}{\internalenergy@symbol^{\excess,t}}
      \renewcommand*{\HEt}{\enthalpy@symbol^{\excess,t}}
      \renewcommand*{\FEt}{\Helmholtz@symbol^{\excess,t}}
      \renewcommand*{\GEt}{\Gibbs@symbol^{\excess,t}}
      \renewcommand*{\VEt}{\volume@symbol^{\excess,t}}
      \renewcommand*{\SEt}{\entropy@symbol^{\excess,t}}
    }
}
```

The next two options choose whether variables held constant are subscripted (the default) or placed next to the function. The `nosubscripts` option currently requires that the variable being changed is the first one in the argument list. The difference is

$$\left(\frac{\partial U}{\partial S}\right)_V \quad \text{versus} \quad \left(\frac{\partial U(S,V)}{\partial S}\right)$$

for `subscripts` and `nosubscripts`, respectively.

```
\newif\if@subscripted
\@subscriptedtrue
\DeclareOption{subscripts}{\@subscriptedtrue}
\DeclareOption{nosubscripts}{\@subscriptedfalse}
```

These options change how `\Partial` and friends render derivatives. The default is parentheses, but other options include brackets, braces, a vertical bar on the right side, or plain (undecorated) derivatives.

```
\DeclareOption{parentheses}{}
\DeclareOption{brackets}{%
  \renewcommand*{\PartialOpen}{[}%
  \renewcommand*{\PartialClose}{]}%
  \renewcommand*{\PartialEmptyClose}{]}%
}
\DeclareOption{braces}{%
  \renewcommand*{\PartialOpen}{\{}%
  \renewcommand*{\PartialClose}{\}}%
  \renewcommand*{\PartialEmptyClose}{\}}%
}
\DeclareOption{bar}{%
  \renewcommand*{\PartialOpen}{.}%
  \renewcommand*{\PartialClose}{\rvert}%
  \renewcommand*{\PartialEmptyClose}{.}%
}
\DeclareOption{plain-derivatives}{% This implies dU(S,V,N)/dS notation
  \renewcommand*{\PartialOpen}{.}%
  \renewcommand*{\PartialClose}{.}%
  \renewcommand*{\PartialEmptyClose}{.}%
  \ExecuteOptions{nosubscripts}%
}
```

4.3 The Number of Moles Macros

\ncomponents

We define the number of components, default C, for use in the "all moles" and related macros.

```
\newcommand*{\ncomponents}{C}
```

\allNs \allNsbut \allMs \allmus \allMsbut \allmusbut

Several macros define a shorthand for "moles of all species" (\allNs) and "moles of all species except" (\allNsbut), as well as similar quantities for masses (\allMs, \allMsbut) and chemical potentials (\allmus, \allmusbut), which occur frequently in mixture thermodynamics. The default is for \allNs to become $\vec{n}$ and allNsbut{i} to become $n_{j\neq i}$. The optional argument changes which index (default: j) to use in the left side of the inequality.[2] Essentially identical commands are defined for chemical potentials and masses: \allmus and \allmusbut and \allMs and \allMsbut, respectively.

```
\newcommand*{\allNs}{\allcomponents{\Nt}}
\newcommand*{\allXs}{\allcomponents{x}}
\newcommand*{\allYs}{\allcomponents{y}}
\newcommand*{\allmus}{\allcomponents{\mu}}
\newcommand*{\allMs}{\allcomponents{m}}
\newcommand*{\allWs}{\allcomponents{w}}%
```

\allXs \allYs \allWs \allWsbut \allXsbut \allYsbut

Similar commands are defined for mole fractions (\allXs, \allYs, etc.), but these assume the last mole fraction is *not* one of the variables—that is, \allXsbut and \allYsbut assume the argument *and* \ncomponents are held constant. For example,

```
\[ \Partial{\Gm}{T}{P,\allXs} = -\Sm \qquad
   \Partial{\Gm}{x_i}{T,P,\allXsbut{i}} \neq \Gpm_i \]
```

yields

$$\left(\frac{\partial G}{\partial T}\right)_{P,\vec{x}} = -S \qquad \left(\frac{\partial G}{\partial x_i}\right)_{T,P,x_{j\neq i,C}} \neq \overline{G}_i.$$

```
\newcommand*{\allNsbut}[2][j]{\allbut[#1]{#2}{\Nt}}
\newcommand*{\allXsbut}[2][j]{\allbutlastand[#1]{#2}{x}}
\newcommand*{\allYsbut}[2][j]{\allbutlastand[#1]{#2}{y}}
\newcommand*{\allmusbut}[2][j]{\allbut[#1]{#2}{\mu}}
\newcommand*{\allMsbut}[2][j]{\allbut[#1]{#2}{m}}
\newcommand*{\allWsbut}[2][j]{\allbutlastand[#1]{#2}{w}}%
```

\allbutlastand \allbut \allcomponents

The \allcomponents, \allbut, and \allbutlastand macros can be used to define new entities; say, if you want to use z_i as a mole fraction, then use

```
\newcommand*{\allZsbut}[2][j]{\allbutlastand[#1]{#2}{z}}
```

Similarly, something meaning the concentrations of every species could be defined via

```
\newcommand*{\allCs}{\allcomponents{C}}
```

```
\newcommand*{\allcomponents}[1]{\vec{#1}}
\newcommand*{\allbut}[3][j]{%
  \def\tmp@arg{#2}%
  \def\tmp@@arg{#1}%
```

[2] The index j is automatically replaced with k if the user issues \allNsbut{j}.

```
  \ifx\tmp@arg\tmp@@arg
    {#3}_{k\neq #2}%
  \else
    {#3}_{#1\neq #2}%
  \fi
}
\newcommand*{\allbutlastand}[3][j]{%
  \edef\tmp@arg{#1}%
  \edef\tmp@@arg{#2}%
  \ifx\tmp@@arg\ncomponents
    {#3}_{#1 \neq #2}%
  \else
    \ifx\tmp@arg\tmp@@arg
      {#3}_{k \neq #2,\ncomponents}%
    \else
      {#3}_{#1 \neq #2,\ncomponents}%
    \fi
  \fi
}
```

We then define two package options that change how to render `\allNs` and friends.

```
\DeclareOption{moles-index}{}
\DeclareOption{moles-range}{%
  \renewcommand*{\allcomponents}[1]{{#1}_1,\dots,{#1}_{\ncomponents}}
  \renewcommand*{\allbut}[3][j]{%
    \def\@one{1}%
    \edef\tmp@arg{#2}%
    \ifx\tmp@arg\@one
      {#3}_2,\dots,{#3}_{\ncomponents}%
    \else
      \ifx\tmp@arg\ncomponents
        {#3}_1,\dots,{#3}_{\ncomponents-1}%
      \else
        {#3}_1,\dots,[{#3}_{#2}],\dots,{#3}_{\ncomponents}%
      \fi
    \fi
  }
  \renewcommand*{\allbutlastand}[3][j]{%
    \def\@one{1}%
    \edef\tmp@arg{#2}%
    \edef\@ncminusone{\ncomponents-1}%
    \ifx\tmp@arg\@one
      {#3}_2,\dots,{#3}_{\ncomponents-1}%
    \else
      \ifx\tmp@arg\@ncminusone
        {#3}_1,\dots,{#3}_{\ncomponents-2}%
      \else
        \ifx\tmp@arg\ncomponents
          {#3}_1,\dots,{#3}_{\ncomponents-1}%
      \else
          {#3}_1,\dots,[{#3}_{#2}],\dots,{#3}_{\ncomponents}%
        \fi
      \fi
    \fi
```

```
  }
}
```

The remaining options define textbook-specific notation.

```
\DeclareOption{Bejan}{
  \ExecuteOptions{EUFGHAN,intensive-lowercase,delta}
  \let\@specific\MakeLowercase
  \def\@intensive#1{\MakeLowercase{\bar #1}}
  \AtEndOfPackage{
    \DeclareSubscrSymbol{cV}{\bar\heatcapacitysymbol}{v}
    \DeclareSubscrSymbol{cVs}{\heatcapacitysymbol}{v}
    \DeclareSubscrSymbol{cVt}{\Nt\heatcapacitysymbol}{v}
    \renewcommand*{\expansivitysymbol}{\beta}
    \renewcommand*{\ncomponents}{n}
  }
}
\DeclareOption{CBK}{
  \ExecuteOptions{EUAGHAN,intensive-lowercase}
  \AtEndOfPackage{
    \let\thermo@overline\widetilde
    \let\@specific\MakeLowercase
    \renewcommand*{\@intensive}[1]{\MakeLowercase{\bar{#1}}}
    \renewcommand*{\pressure@symbol}{p}
    \renewcommand*{\Deltarxn}[1]{#1_R}
    \renewcommand*{\compressibilitysymbol}{\beta}
    \renewcommand*{\expansivitysymbol}{\alpha}
  }
}
\DeclareOption{ElliottLira}{
  \AtEndOfPackage{
    \renewcommand*{\allcomponents}[1]{#1}
    \renewcommand*{\Deltarxn}[1]{\Delta #1}
  }
}
\DeclareOption{Koretsky}{
  \ExecuteOptions{EUAGHAn,brackets,intensive-lowercase,delta}
  \AtEndOfPackage{
    \renewcommand*{\partialmolar}[1]{\gdef\pm@symbol{#1}\generic@pm}
    \renewcommand*{\expansivitysymbol}{\beta}
    \renewcommand*{\IS}{{\text{ideal}}}
    \renewcommand*{\residual}{{\text{dep}}}
    \renewcommand*{\IG}{{\text{ideal}}}
    \renewcommand*{\IGM}{{\text{ideal}}}
    \renewcommand*{\Deltafus}[1]{\Delta{#1}_\fusion}
    \renewcommand*{\Deltasub}[1]{\Delta{#1}_\sublimation}
    \renewcommand*{\Deltavap}[1]{\Delta{#1}_\vaporization}
    \renewcommand*{\Henryrat}{{\mathcal{H}}}
    \renewcommand*{\gammarat}{\gamma^\text{Henry's}}
    \let\phipure\varphi
    \renewcommand*{\phimix}{\hat\varphi}
    \renewcommand*{\phisat}{\varphi^\sat}
    \let\Delta@fus@sym\relax
    \DeclareSubscrSymbol{@Deltafus}{\Delta@fus@sym}{\fusion}
    \renewcommand*{\Deltafus}[1]{\def\Delta@fus@sym{\Delta #1}\@Deltafus}
```

```
    \let\Delta@vap@sym\relax
    \DeclareSubscrSymbol{@Deltavap}{\Delta@vap@sym}{\vaporization}
    \renewcommand*{\Deltasub}[1]{\def\Delta@sub@sym{\Delta #1}\@Deltasub}
    \let\Delta@sub@sym\relax
    \DeclareSubscrSymbol{@Deltasub}{\Delta@sub@sym}{\sublimation}
    \renewcommand*{\Deltasub}[1]{\def\Delta@sub@sym{\Delta #1}\@Deltasub}
  }
}
\DeclareOption{MSBB}{
  \ExecuteOptions{EUFGHAn,intensive-lowercase,delta}
  \AtEndOfPackage{
    \renewcommand*{\IGM}{\ast}
    \renewcommand*{\IG}{\ast}
    \renewcommand*{\expansivitysymbol}{\beta}
    \renewcommand*{\allcomponents}[1]{#1}
    \renewcommand*{\allbut}[3][j]{%
      \def\tmp@arg{#2}%
      \def\tmp@@arg{#1}%
      \ifx\tmp@arg\tmp@@arg
        {#3}_{k}%
      \else
        {#3}_{#1}%
      \fi
    }
    \let\Helmholtz@symbol\psi
    \renewcommand*{\Ft}{\Psi}
    \def\@intensive#1{\MakeLowercase{\thermo@overline #1}}
    \let\@specific\MakeLowercase
    \renewcommand*{\partialmolar}[1]{\gdef\pm@symbol{#1}\generic@pm}
    \renewcommand*{\fmix}{\bar f}
    \renewcommand*{\phimix}{\bar\phi}
    \renewcommand*{\pressure@symbol}{p}
  }
}
\DeclareOption{Prausnitz}{
  \ExecuteOptions{intensive-lowercase}
  \AtEndOfPackage{
    \renewcommand*{\fmix}{f}
    \let\phimix\phi
    \renewcommand*{\fsat}{\fpure^\sat}
    \DeclareSubscrSymbol{fpure}{f}{{\text{pure}}}
    \DeclareSubscrSymbol{phipure}{\phi}{{\text{pure}}}
    \renewcommand*{\residual}{{{\mathcal{R}}}}
    \renewcommand*{\allcomponents}[1]{{#1}_i}
    \renewcommand*{\allbut}[3][j]{%
      \def\tmp@arg{#2}%
      \def\tmp@@arg{#1}%
      \ifx\tmp@arg\tmp@@arg
        {#3}_{k}%
      \else
        {#3}_{#1}%
      \fi
    }
  }
```

```
}
\DeclareOption{Sandler}{
  \ExecuteOptions{EUAGHaN,extensive-plain}
  \AtEndOfPackage{
    \renewcommand*{\sat}{{\text{vap}}}
    \renewcommand*{\excess}{{\text{{ex}}}}
    \renewcommand*{\residual}{{\text{r}}}
    \renewcommand*{\ncomponents}{\mathcal{C}}
    \renewcommand*{\fmix}{\bar f}
    \renewcommand*{\fstd}{\bar f^\circ}
    \renewcommand*{\phimix}{\bar\phi}
    \renewcommand*{\allcomponents}[1]{\thermo@underline{#1}}
    \renewcommand*{\IG}{{\text{IG}}}
    \renewcommand*{\IGM}{{\text{IGM}}}
    \renewcommand*{\IS}{{\text{IM}}}
    \renewcommand*{\Deltamix}[1]{\Delta_\mixing #1}
    \renewcommand*{\Deltarxn}[1]{\Delta_\reaction #1}
    \renewcommand*{\Deltasub}[1]{\Delta_\sublimation #1}
    \renewcommand*{\Deltafus}[1]{\Delta_\fusion #1}
    \renewcommand*{\Deltavap}[1]{\Delta_\vaporization #1}
    \renewcommand*{\Henryrat}{H}
    \DeclareSubscrSymbol{cV}{\heatcapacitysymbol}{\volume@symbol}
    \DeclareSubscrSymbol{cP}{\heatcapacitysymbol}{\pressure@symbol}
    \DeclareSubscrSymbol{cVt}{\Nt\heatcapacitysymbol}{\volume@symbol}
    \DeclareSubscrSymbol{cPt}{\Nt\heatcapacitysymbol}{\pressure@symbol}
  }
}
\DeclareOption{SVNAS}{
  \ExecuteOptions{extensive-superscript}
  \AtEndOfPackage{
    \renewcommand*{\allcomponents}[1]{#1}
    \renewcommand*{\allbut}[3][j]{%
      \def\tmp@arg{#2}%
      \def\tmp@@arg{#1}%
      \ifx\tmp@arg\tmp@@arg
        {#3}_{k}%
      \else
        {#3}_{#1}%
      \fi
    }
    \renewcommand*{\IG}{{ig}}
    \renewcommand*{\IGM}{{ig}}
    \renewcommand*{\IS}{{id}}
    \renewcommand*{\expansivitysymbol}{\beta}
    \renewcommand*{\Deltarxn}[1]{\Delta #1}
    \DeclareSubscrSymbol{cVt}{\Nt\heatcapacitysymbol}{\volume@symbol}
    \DeclareSubscrSymbol{cPt}{\Nt\heatcapacitysymbol}{\pressure@symbol}
  }
}
\DeclareOption{TesterModell}{
  \ExecuteOptions{EUAGHaN,delta}
  \AtEndOfPackage{
    \renewcommand*{\ncomponents}{n}
    \renewcommand*{\allcomponents}[1]{{#1}_i}
```

```
    \renewcommand*{\allbut}[3][j]{%
      \def\tmp@arg{#2}%
      \def\tmp@@arg{#1}%
      \ifx\tmp@arg\tmp@@arg
        {#3}_k\relax[#2]
      \else
        {#3}_{#1}\relax[#2]
      \fi
    }
    \renewcommand*{\allbutlastand}[3][j]{%
      \edef\tmp@arg{#1}%
      \edef\tmp@@arg{#2}%
      \ifx\tmp@@arg\ncomponents
        {#3}_{#1}\relax[#2]%
      \else
        \ifx\tmp@arg\tmp@@arg
          {#3}\relax[#2,\ncomponents]%
        \else
          {#3}\relax[#2,\ncomponents]%
        \fi
      \fi
    }
    \renewcommand*{\IG}{{ig}}
% Tester & Modell never use "igm" anywhere, and I only found ig once
    \renewcommand*{\IGM}{{igm}}
    \renewcommand*{\IS}{{ID}}
    \renewcommand*{\excess}{{EX}}%
    \renewcommand*{\reaction}{{rx}}
    \renewcommand*{\Henryrat}{f^{\ast\ast}}
    \renewcommand*{\Henrymol}{f^\ast}
    \renewcommand*{\gammarat}{\gamma^{\ast\ast}}
    \renewcommand*{\gammamol}{\gamma^\ast}
  }
}
\DeclareOption{Thompson}{
  \ExecuteOptions{EUAGHAn,delta}
  \AtEndOfPackage{
    \renewcommand*{\excess}{{EX}}
    \renewcommand*{\residual}{{R}}
    \renewcommand*{\allcomponents}[1]{{#1}_j}
    \renewcommand*{\ncomponents}{c}
    \renewcommand*{\IS}{{IS}}
    \renewcommand*{\IG}{{IG}}
    \let\IGM\IG
    \def\@fpure_#1{f_{#1}\@ifnextchar^{}{^\bullet}}
    \renewcommand*{\fpure}{\@ifnextchar_{\@fpure}{f}}
    \def\@phi@pure_#1{\phi_{#1}\@ifnextchar^{}{^\bullet}}
    \renewcommand*{\phipure}{\@ifnextchar_{\@phi@pure}{\phi}}
    \renewcommand*{\mixing}{{MIX}}
    \renewcommand*{\Deltamix}[1]{\Delta_\mixing #1}
    \renewcommand*{\Deltarxn}[1]{\Delta #1}
    \def\@@intensive_#1{_{#1}\@ifnextchar^{}{^\bullet}}
    \newcommand*{\@intensive}[1]{#1\@ifnextchar_{\@@intensive}{}}
    \renewcommand*{\allbut}[3][j]{%
```

```
      \def\tmp@arg{#2}%
      \def\tmp@@arg{#1}%
      \ifx\tmp@arg\tmp@@arg
        {#3}_{k}\neq{#3}_{#2}%
      \else
        {#3}_{#1}\neq{#3}_{#2}%
      \fi
    }
  }
}
```

We execute the default options below.

```
\ExecuteOptions{EUAGHan,subscripts,parentheses,intensive-plain,moles-index}
\ProcessOptions
```

4.4 Commands for Partial Derivatives

The `\Partial` command and its second-order siblings are defined as below. They typeset partial derivatives of the first argument with respect to the second (and third, in the case of mixed second partial derivatives) arguments, holding the last argument constant.

The starred forms adjust the spacing after the partial derivative so the trailing binary operator (assumed to be the same width as an equals sign) overhangs the variables held constant. We thus set `operator@width` to be *just* greater than the width of an equals sign.

```
\newlength{\Partial@const@width}
\newlength{\operator@width}
\settowidth{\operator@width}{=}
\newlength{\adjust@width}
\setlength{\adjust@width}{0.1\operator@width}
\addtolength{\operator@width}{\adjust@width}
```

`\Partial*` `\Partial` The command `\Partial` and its friends drastically simplify the creation of partial derivatives. The command `\Partial*` is the same as `\Partial` except that it adjusts the spacing so the (presumably) binary operator that follows it slightly overlaps the subscripts.

```
\def\Partial@start{\left\PartialOpen}
\def\Partial@end{\right\PartialClose}
\def\Partial@empty@end{\right\PartialEmptyClose}
\def\Partial{\@ifstar\Partial@star\Partial@nostar}
%^^A Several ifs to use later (breaks it if these definitions are inside)
\newif\iftwo@has@Nt
\two@has@Ntfalse
\newif\iftwo@has@xory
\two@has@xoryfalse
\newif\ifthree@has@Nt
\three@has@Ntfalse
\newif\ifthree@has@xory
\three@has@xoryfalse
\AtEndOfPackage{%
  \edef\thermo@Nt{\Nt}%
  \edef\thermo@x{x}%
```

```
  \edef\thermo@y{y}%
}
\def\thermo@parse@two#1\relax{%
  \@tfor\@i:=#1\do{%
    \edef\@@i{\@i}%
    \ifx\@@i\thermo@Nt
      \two@has@Nttrue
      \@break@tfor
    \fi
    \ifx\@@i\thermo@x
      \two@has@xorytrue
      \@break@tfor
    \fi
    \ifx\@@i\thermo@y
      \two@has@xorytrue
      \@break@tfor
    \fi
  }
}
\newcommand*{\Partial@nostar}[3]{%
  \def\tmp@arg{#3}%
  \if@subscripted
    % Handles situation of empty variables held constant
    \ifx\tmp@arg\@empty
      \ensuremath{\Partial@start\frac{\partial #1}
                                      {\partial #2}\Partial@empty@end}%
    \else
      \ensuremath{\Partial@start\frac{\partial #1}
                                      {\partial #2}\Partial@end_{#3}}%
    \fi
  \else
    % If not subscripted, we put the arguments (if any) in the derivative
    \ifx\tmp@arg\@empty
      \ensuremath{\Partial@start\frac{\partial #1}
                                      {\partial #2}\Partial@empty@end}%
    \else
      % Check for whether #3 contains \allNsbut{i}/etc. and #2 is \Nt_i/etc.
      \thermo@parse@two#2\relax
      \iftwo@has@Nt
        \renewcommand*{\allbut}[3][j]{\allcomponents{##3}}%
        \ensuremath{\Partial@start\frac{\partial #1(#3)}
                                        {\partial #2}\Partial@end}%
      \else
        \iftwo@has@xory
          \renewcommand*{\allbutlastand}[3][j]{\allbut{\ncomponents}{##3}}%
          \ensuremath{\Partial@start\frac{\partial #1(#3)}
                                          {\partial #2}\Partial@end}%
        \else
          \ensuremath{\Partial@start\frac{\partial #1(#2,#3)}
                                          {\partial #2}\Partial@end}%
        \fi
      \fi
    \fi
  \fi
```

```
}
\newcommand*{\Partial@star}[3]{%
  \settowidth{\Partial@const@width}{\ensuremath{#3}}%
  \addtolength{\Partial@const@width}{-0.15\Partial@const@width}%
  \Partial@nostar{#1}{#2}{#3}%
  \if@subscripted
    \ifdim\operator@width<\Partial@const@width
      \kern -\operator@width
    \else
      \kern -\Partial@const@width
    \fi
  \fi
}
```

\PartialBigg
\PartialBigg*

The \PartialBigg macro (and its starred form) replace the \left and \right commands in \Partial with amsmath's \Biggl and \Biggr variants. The starred form is inherited from \Partial without modification.

```
\def\PartialBigg{%
  \def\Partial@start{\Biggl\PartialOpen}%
  \def\Partial@end{\Biggr\PartialClose}%
  \def\Partial@empty@end{\Biggr\PartialClose}%
  \Partial%
}
```

\Partialbigg
\Partialbigg*

The \Partialbigg macro does the same thing as \PartialBigg, except using amsmath's \biggl/\biggr variants.

```
\def\Partialbigg{%
  \def\Partial@start{\biggl\PartialOpen}%
  \def\Partial@end{\biggr\PartialClose}%
  \def\Partial@empty@end{\biggr\PartialClose}%
  \Partial%
}
```

\PartialSecond
\PartialSecond*

The second partial derivatives are defined similarly to \Partial.

```
\def\PartialSecond{\@ifstar\PartialSecond@star\PartialSecond@nostar}
\newcommand*{\PartialSecond@nostar}[3]{%
  \def\tmp@arg{#3}%
  \if@subscripted
  % Handles situation of empty variables held constant
    \ifx\tmp@arg\@empty
        \ensuremath{\Partial@start\frac{\partial^2 #1}
                                       {\partial #2^2}\Partial@empty@end}%
    \else
        \ensuremath{\Partial@start\frac{\partial^2 #1}
                                       {\partial #2^2}\Partial@end_{#3}}%
    \fi
  \else
    \ifx\tmp@arg\@empty
        \ensuremath{\Partial@start\frac{\partial^2 #1}
                                       {\partial #2^2}\Partial@empty@end}%
    \else
      % Check for whether #3 contains \allNsbut{i}/etc. and #2 is \Nt_i/etc.
      \thermo@parse@two#2\relax
```

```
      \iftwo@has@Nt
        \renewcommand*{\allbut}[3][j]{\allcomponents{##3}}%
        \ensuremath{\Partial@start\frac{\partial^2 #1(#3)}
                                        {\partial #2^2}\Partial@end}%
      \else
        \iftwo@has@xory
          \renewcommand*{\allbutlastand}[3][j]{\allbut{\ncomponents}{##3}}%
          \ensuremath{\Partial@start\frac{\partial^2 #1(#3)}
                                          {\partial #2^2}\Partial@end}%
        \else
          \ensuremath{\Partial@start\frac{\partial^2 #1(#2,#3)}
                                          {\partial {#2}^2}\Partial@end}%
        \fi
      \fi
    \fi
  \fi
}
\newcommand*{\PartialSecond@star}[3]{%
    \settowidth{\Partial@const@width}{\ensuremath{#3}}%
    \addtolength{\Partial@const@width}{-0.15\Partial@const@width}%
    \PartialSecond@nostar{#1}{#2}{#3}%
    \if@subscripted
      \ifdim\operator@width<\Partial@const@width
        \kern -\operator@width
      \else
        \kern -\Partial@const@width
      \fi
    \fi
}
```

`\PartialSecondBigg` `\PartialSecondBigg*` The `\PartialSecondBigg` macro and its starred variant replace `\left` and `\right` with amsmath's `\Biggl` and `\Biggr`.

```
\def\PartialSecondBigg{%
  \def\Partial@start{\expandafter\Biggl\PartialOpen}%
  \def\Partial@end{\expandafter\Biggr\PartialClose}%
  \PartialSecond%
}
```

`\PartialSecondbigg` `\PartialSecondbigg*` The `\PartialSecondbigg` macro and its starred variant replace `\left` and `\right` with amsmath's `\biggl` and `\biggr`.

```
\def\PartialSecondbigg{%
  \def\Partial@start{\expandafter\biggl\PartialOpen}%
  \def\Partial@end{\expandafter\biggr\PartialClose}%
  \PartialSecond%
}
```

`\PartialMixSecond` `\PartialMixSecond*` The macro `\PartialMixSecond` takes an extra argument, but is otherwise the same as its same-variable cousin.

```
\def\PartialMixSecond{\@ifstar\PartialMixSecond@star\PartialMixSecond@nostar}
\newcommand*{\PartialMixSecond@nostar}[4]{%
  \def\tmp@arg{#4}%
  \if@subscripted
  % Handles situation of empty variables held constant
```

```
    \ifx\tmp@arg\@empty
      \ensuremath{\Partial@start\frac{\partial^2 #1}
                           {\partial #2\partial#3}\Partial@empty@end}%
    \else
      \ensuremath{\Partial@start\frac{\partial^2 #1}
                           {\partial #2\partial#3}\Partial@end_{#4}}%
    \fi
  \else
    \ifx\tmp@arg\@empty
      \ensuremath{\Partial@start\frac{\partial^2 #1(#2,#3)}
                           {\partial #2\partial#3}\Partial@empty@end}%
    \else
      % This time, we check whether #2 OR #3 are \Nt_i/etc.
      \thermo@parse@two#3\relax
      \iftwo@has@Nt\three@has@Nttrue\fi
      \iftwo@has@xory\three@has@xorytrue\fi
      \two@has@Ntfalse
      \two@has@xoryfalse
      \thermo@parse@two#2\relax
      \iftwo@has@Nt
        \renewcommand*{\allbut}[3][j]{\allcomponents{##3}}%
        \ensuremath{\Partial@start\frac{\partial^2 #1(#3,#4)}
                                       {\partial #2\partial#3}\Partial@end}%
      \else
        \iftwo@has@xory
          \renewcommand*{\allbutlastand}[3][j]{\allbut{\ncomponents}{##3}}%
          \ensuremath{\Partial@start\frac{\partial^2 #1(#3,#4)}
                                         {\partial #2\partial#3}\Partial@end}%
        \else
          \ifthree@has@Nt
            \renewcommand*{\allbut}[3][j]{\allcomponents{##3}}%
            \ensuremath{\Partial@start\frac{\partial^2 #1(#2,#4)}
                                       {\partial #2\partial#3}\Partial@end}%
          \else
            \ifthree@has@xory
              \PackageWarning{thermodynamics}{GOT HERE}
              %\renewcommand*{\allbutlastand}[3][j]{\allbut{\ncomponents}{##3}}%
              \renewcommand*{\allbutlastand}[3][j]{SCREW YOU}%
              \ensuremath{\Partial@start\frac{\partial^2 #1(#2,#4)}
                                         {\partial #2\partial#3}\Partial@end}%
            \else
              \ensuremath{\Partial@start\frac{\partial^2 #1(#2,#3,#4)}
                                         {\partial #2\partial#3}\Partial@end}%
            \fi
          \fi
        \fi
      \fi
    \fi
  \fi
}
\newcommand*{\PartialMixSecond@star}[4]{%
    \settowidth{\Partial@const@width}{\ensuremath{#4}}%
    \addtolength{\Partial@const@width}{-0.25\Partial@const@width}%
    \PartialMixSecond@nostar{#1}{#2}{#3}{#4}%
```

```
    \if@subscripted
      \ifdim\operator@width<\Partial@const@width
        \kern -\operator@width
      \else
        \kern -\Partial@const@width
      \fi
    \fi
}
```

\PartialMixSecondBigg
\PartialMixSecondBigg*
\PartialMixSecondbigg
\PartialMixSecondbigg*

The macro \PartialMixSecondBigg and its starred form are analogous to \PartialBigg and \PartialSecondBigg. \PartialMixSecondbigg and its starred form are similarly analogous to \Partialbigg and \PartialSecondbigg.

```
\def\PartialMixSecondBigg{%
  \def\Partial@start{\expandafter\Biggl\PartialOpen}%
  \def\Partial@end{\expandafter\Biggr\PartialClose}%
  \PartialMixSecond%
}
\def\PartialMixSecondbigg{%
  \def\Partial@start{\expandafter\biggl\PartialOpen}%
  \def\Partial@end{\expandafter\biggr\PartialClose}%
  \PartialMixSecond%
}
```

If the user does not load the amsmath package, we will not have access to \Biggl, \Biggr, \biggl, and \biggr, so we revert them back to the ordinary \left and \right versions and warn the user.

```
\AtBeginDocument{%
  \@ifpackageloaded{amsmath}{}{%
      \PackageWarningNoLine{thermodynamics}
        {Package amsmath not loaded; load to make PartialBigg and friends
            work correctly}%
      \let\PartialBigg\Partial
      \let\Partialbigg\Partial
      \let\PartialSecondBigg\PartialSecond
      \let\PartialSecondbigg\PartialSecond
      \let\PartialMixSecondBigg\PartialMixSecond
      \let\PartialMixSecondbigg\PartialMixSecond
  }%
}
```

4.5 Local Override of Delimiters

thermoparentheses
thermobrackets
thermobraces
thermobar
thermoplain

We define seven environments that *locally* override the delimiters on partial derivatives generated with \Partial and friends and/or the subscript notation for partial derivatives.

```
\newenvironment*{thermoparentheses}{
  \renewcommand*{\PartialOpen}{(}%
  \renewcommand*{\PartialClose}{)}%
  \renewcommand*{\PartialEmptyClose}{)}%
}{}
\newenvironment*{thermobrackets}{%
  \renewcommand*{\PartialOpen}{[}%
  \renewcommand*{\PartialClose}{]}%
```

```
  \renewcommand*{\PartialEmptyClose}{]}%
}{}
\newenvironment*{thermobraces}{%
  \renewcommand*{\PartialOpen}{\{}%
  \renewcommand*{\PartialClose}{\}}%
  \renewcommand*{\PartialEmptyClose}{\}}%
}{}
\newenvironment*{thermobar}{%
  \renewcommand*{\PartialOpen}{.}%
  \renewcommand*{\PartialClose}{\rvert}%
  \renewcommand*{\PartialEmptyClose}{.}%
}{}
\newenvironment*{thermoplain}{%
  \renewcommand*{\PartialOpen}{.}%
  \renewcommand*{\PartialClose}{.}%
  \renewcommand*{\PartialEmptyClose}{.}%
  \@subscriptedfalse
}{}
\newenvironment*{thermoNOsubscripts}{\@subscriptedfalse}{}
\newenvironment*{thermosubscripts}{\@subscriptedtrue}{}
```

4.6 User-Interface Macros to Define Symbols

\DeclareSubscrSymbol

First, we define a (messy!) command that serves to create "subscripted" symbols; for example, typing `\cP_i` should yield $C_{P,i}$ rather than C_{Pi}, C_{P_i}, or C_{P_i}. Superscripts are also handled properly and can be in either order.

```
\newcommand*{\DeclareSubscrSymbol}[3]{%
  \expandafter\def\csname #1\endcsname{%
    \@ifnextchar^{\csname @#1\endcsname}
                 {\@ifnextchar_{\csname @@#1\endcsname}{{#2}_{#3}}}
  }
  \expandafter\def\csname @#1\endcsname ##1##2{%
    {#2}^{##2}\@ifnextchar_{\csname @@@#1\endcsname}{_{#3}}%
  }
  \expandafter\def\csname @@#1\endcsname ##1##2{{#2}_{#3,##2}}
  \expandafter\def\csname @@@#1\endcsname ##1##2{_{#3,##2}}
}
```

\heatcapacitysymbol
\compressibilitysymbol
\expansivitysymbol

Now we define symbols for the heat capacities, compressibilities, and so forth.

```
\newcommand*{\heatcapacitysymbol}{C}
\newcommand*{\compressibilitysymbol}{\kappa}
\newcommand*{\expansivitysymbol}{\alpha}
```

\cV
\cP

The heat capacities are molar by default; we also declare extensive and specific heat capacities. The heat capacities themselves are defined to be "smart": `\cV_i` will recognize the subscript appropriately and render $C_{V,i}$ rather than C_{Vi} or some other unintended symbol. Superscripts are also handled appropriately and can be in either order.

```
\DeclareSubscrSymbol{cV}{\@intensive\heatcapacitysymbol}{\volume@symbol}
\DeclareSubscrSymbol{cP}{\@intensive\heatcapacitysymbol}{\pressure@symbol}
```

\cVt
\cPt

We also introduce *extensive* (rather than molar) heat capacity macros.

```
\DeclareSubscrSymbol{cVt}{\@extensive\heatcapacitysymbol}{\volume@symbol}
\DeclareSubscrSymbol{cPt}{\@extensive\heatcapacitysymbol}{\pressure@symbol}
```

\cPs
\cVs

`\cPs` and `\cVs` are the specific heat capacities.

```
\DeclareSubscrSymbol{cVs}{\@specific\heatcapacitysymbol}{\volume@symbol}
\DeclareSubscrSymbol{cPs}{\@specific\heatcapacitysymbol}{\pressure@symbol}
```

\kappaT
\kappaS

The isothermal and adiabatic compressibilities are defined similarly, but those do not have extensive versions for obvious reasons.

```
\DeclareSubscrSymbol{kappaT}{\compressibilitysymbol}{\temperature@symbol}
\DeclareSubscrSymbol{kappaS}{\compressibilitysymbol}{\entropy@symbol}
```

\alphaP
\alphaS

The macro `\alphaP` is intended to refer to the isobaric volume expansivity, while `\alphaS` is the isentropic volume expansivity.

```
\DeclareSubscrSymbol{alphaP}{\expansivitysymbol}{\pressure@symbol}
\DeclareSubscrSymbol{alphaS}{\expansivitysymbol}{\entropy@symbol}
```

\Psat
\Pvap

The `\Psat` macro (and its clone, the `\Pvap` macro) should be used for the saturation pressure.

\phisat
\fsat
\sat

Similarly, a `\phisat` macro typesets the fugacity coefficient at saturation. The `\fsat` macro similarly renders the fugacity at saturation. Other saturation properties should use `M^\sat` or similar, preferably by defining another macro such as `\Gmsat`.

```
\newcommand*{\sat}{{\text{sat}}}
\newcommand*{\Psat}{P^\sat}
\let\Pvap\Psat
\newcommand*{\phisat}{\phi^\sat}
\newcommand*{\fsat}{f^\sat}
```

std
Pstd
fstd

The `\std` macro denotes standard properties. `\Pstd` and `\fstd` are defined for convenience and for consistency across textbooks.

```
\let\std\circ
\newcommand*{\Pstd}{P^\std}
\newcommand*{\fstd}{f^\std}
```

\Deltamix
\mixing

Mixing properties, such as ΔG_{mix}, should be accessed using `\Deltamix\Gm` and similar constructions—this construct will typeset as $\Delta_{MIX}G$ using the `Thompson` package option, for example, and as $\Delta_{\text{mix}}G$ using the `Sandler` package option.

```
\newcommand*{\mixing}{{\text{mix}}}
\newcommand*{\Deltamix}[1]{\Delta{#1}_\mixing}
```

\Deltafus
\fusion
\Deltavap
\vaporization
\Deltasub
\sublimation
\Deltarxn
\reaction

Similar entities for property changes on fusion, reaction, sublimation, and vaporization are defined. The heat of reaction is handled differently if `\cP` is its argument: one must typeset $\Delta C_{P,\text{rxn}}$ and $\Delta C^{\circ}_{P,\text{rxn}}$ rather than $\Delta C_{P\text{rxn}}$ and $\Delta C^{\circ}_{P\text{rxn}}$.

```
\newcommand*{\fusion}{{\text{fus}}}
\newcommand*{\reaction}{{\text{rxn}}}
\newcommand*{\sublimation}{{\text{sub}}}
\newcommand*{\vaporization}{{\text{vap}}}
\newcommand*{\formation}{f}
\newcommand*{\Deltafus}[1]{\Delta{#1}^\fusion}
\newcommand*{\Deltasub}[1]{\Delta{#1}^\sublimation}
\newcommand*{\Deltavap}[1]{\Delta{#1}^\vaporization}
```

```
% TODO It would be nice not to have to handle \cP in a special way....
\let\Delta@rxn@sym\relax
\DeclareSubscrSymbol{@DeltarxncP}{\Delta@rxn@sym}{\pressure@symbol,\reaction}
\DeclareSubscrSymbol{@Deltarxn}{\Delta@rxn@sym}{\reaction}
\newcommand*{\Deltarxn}[1]{%
    \def\tmp@arg{#1}%
    \def\tmp@@arg{\cP}%
    \ifx\tmp@arg\tmp@@arg\relax
      \def\@command{\def\Delta@rxn@sym{\Delta\@intensive\heatcapacitysymbol}%
                    \@DeltarxncP}
    \else
      \def\@command{\def\Delta@rxn@sym{\Delta{#1}}\@Deltarxn}
    \fi
    \@command
}
\let\Delta@f@sym\relax
\DeclareSubscrSymbol{@Deltaf}{\Delta@f@sym}{f}
\newcommand*{\Deltaf}[1]{\def\Delta@f@sym{\Delta{#1}}\@Deltaf}
```

`\fmix` The `\fmix` command is intended to describe fugacities in mixtures. It renders as $\hat{f}$ by default, and would be used as `\fmix_i` or the like, producing $\hat{f}_i$; some authors like to use $\bar{f}_i$ or just f_i, and this command creates a consistent way to change between such options.

```
\newcommand*{\fmix}{\hat{f}}
```

`\phimix` A similar command, `\phimix`, renders $\hat{\phi}$ by default to represent the fugacity coefficient in the mixture.

```
\newcommand*{\phimix}{\hat\phi}
```

`\fpure` The `\fpure` command is intended to describe fugacities in pure substances. It renders as f by default, and would be used as `\fpure` or `\fpure_i` or the like, producing f_i; some authors like to use $f_{\text{pure},i}$, and others like to use $f_i^\bullet$; this command creates a consistent way to change between the these options. A similar command for ϕ is given, `\phipure`, for fugacity coefficients.

```
\newcommand*{\fpure}{f}
\let\phipure\phi
```

4.7 Partial Molar Quantities

`\partialmolar` Partial molar quantities appear as $\overline{G_i}$ or, for residual properties, $\overline{G_i^R}$, rather than something like $\bar{G}_i$ or $\bar{G}_i^R$, which looks better but is harder to implement for obvious reasons. Their definitions allow them to be used as symbols, something like `\Gpm_i`, `\Gpm^{\IGM}_i`, `\Gpm^{\IGM}_i`, and even `\Gpm_i^{\IGM}`; they can also be treated as commands: `\Gpm{i}` is equivalent to `\Gpm_i` and `\Gpm[\IGM]{i}` is equivalent to `\Gpm_i^{\IGM}`. The macro `\partialmolar` can be used to create an arbitrary partial molar symbol.

```
\let\pm@symbol\relax
\newcommand*{\partialmolar}[1]{\gdef\pm@symbol{#1}\generic@pm}
% if nextchar is _
\def\generic@pm{\@ifnextchar_\@generic@pm\@@generic@pm}
```

```
% then store the argument
\def\@generic@pm#1#2{%
    \gdef\pm@arg{#2}%
    % if nextchar after argument is ^
    \@ifnextchar^\@@@generic@pm\@@@@generic@pm
}
% then print it this way
\def\@@@generic@pm#1#2{\thermo@overline{\pm@symbol^{#2}_{\pm@arg}}}
% else do it this way
\def\@@@@generic@pm{\thermo@overline{\pm@symbol_{\pm@arg}}}
% else, check whether there is also a ^ coming, otherwise assume
% it's [] or {}
\def\@@generic@pm{\@ifnextchar^\@@@@@generic@pm\@pm@bracket@check}
\def\@@@@@generic@pm#1#2#3#4{\thermo@overline{\pm@symbol^{#2}_{#4}}}
\def\@pm@bracket@check{\@ifnextchar[\@pm@bracket@yes\@pm@bracket@no}
\def\@pm@bracket@yes[#1]#2{\thermo@overline{\pm@symbol^{#1}_{#2}}}
\def\@pm@bracket@no#1{\thermo@overline{\pm@symbol_{#1}}}
```

4.8 Symbol Definitions

These macros define the user interface to the symbols for energy, volume, and so forth. We define the "extra" symbols `\Bt` and `\Mt` to represent arbitrary properties.

`\Nt` `\Et` `\Ut` `\Ft` `\Gt` `\Ht` `\Lt` `\At` `\St` `\Vt` `\Qt` `\Wt` `\Mt` `\Bt`

First, the extensive properties.

```
\newcommand*{\Nt}{\mole@symbol}
\newcommand*{\Et}{\@extensive\totalenergy@symbol}
\newcommand*{\Ut}{\@extensive\internalenergy@symbol}
\newcommand*{\Ft}{\@extensive\Helmholtz@symbol}
\newcommand*{\Gt}{\@extensive\Gibbs@symbol}
\newcommand*{\Ht}{\@extensive\enthalpy@symbol}
\newcommand*{\Lt}{\@extensive\Landau@symbol}
\newcommand*{\At}{\@extensive\area@symbol}
\newcommand*{\St}{\@extensive\entropy@symbol}
\newcommand*{\Vt}{\@extensive\volume@symbol}
\newcommand*{\Qt}{\@extensive\heat@symbol}
\newcommand*{\Wt}{\@extensive\work@symbol}
\newcommand*{\Mt}{\@extensive M}
\newcommand*{\Bt}{\@extensive B}
```

`\Em` `\Um` `\Fm` `\Gm` `\Hm` `\Lm` `\Am` `\Sm` `\Vm` `\Qm`

Then the molar properties.

```
\newcommand*{\Em}{\@intensive\totalenergy@symbol}
\newcommand*{\Um}{\@intensive\internalenergy@symbol}
\newcommand*{\Fm}{\@intensive\Helmholtz@symbol}
\newcommand*{\Gm}{\@intensive\Gibbs@symbol}
\newcommand*{\Hm}{\@intensive\enthalpy@symbol}
\newcommand*{\Lm}{\@intensive\Landau@symbol}
\newcommand*{\Am}{\@intensive\area@symbol}
\newcommand*{\Sm}{\@intensive\entropy@symbol}
\newcommand*{\Vm}{\@intensive\volume@symbol}
\newcommand*{\Qm}{\@intensive\heat@symbol}
```

`\Wm` `\Mm` `\Bm`

More molar properties.

```
\newcommand*{\Wm}{\@intensive\work@symbol}
```

```
\newcommand*{\Mm}{\@intensive M}
\newcommand*{\Bm}{\@intensive B}
```

\Es \Us \Fs \Gs \Hs \Ls \As \Ss \Vs \Qs \Ws \Ms \Bs

Now the specific (per unit mass) properties.

```
\newcommand*{\Us}{\@specific\internalenergy@symbol}
\newcommand*{\Es}{\@specific\totalenergy@symbol}
\newcommand*{\Fs}{\@specific\Helmholtz@symbol}
\newcommand*{\Gs}{\@specific\Gibbs@symbol}
\newcommand*{\Hs}{\@specific\enthalpy@symbol}
\newcommand*{\Ls}{\@specific\Landau@symbol}
\newcommand*{\As}{\@specific\area@symbol}
\newcommand*{\Ss}{\@specific\entropy@symbol}
\newcommand*{\Vs}{\@specific\volume@symbol}
\newcommand*{\Qs}{\@specific\heat@symbol}
\newcommand*{\Ws}{\@specific\work@symbol}
\newcommand*{\Ms}{\@specific M}
\newcommand*{\Bs}{\@specific B}
```

\Epm \Upm \Hpm \Fpm \Gpm \Apm \Spm \Vpm \Lpm \Mpm \Bpm

Now the partial molar properties.

```
\newcommand*{\Epm}{\partialmolar{\totalenergy@symbol}}
\newcommand*{\Upm}{\partialmolar{\internalenergy@symbol}}
\newcommand*{\Hpm}{\partialmolar{\enthalpy@symbol}}
\newcommand*{\Fpm}{\partialmolar{\Helmholtz@symbol}}
\newcommand*{\Gpm}{\partialmolar{\Gibbs@symbol}}
\newcommand*{\Apm}{\partialmolar{\area@symbol}}
\newcommand*{\Spm}{\partialmolar{\entropy@symbol}}
\newcommand*{\Vpm}{\partialmolar{\volume@symbol}}
\newcommand*{\Lpm}{\partialmolar{\Omega}}
\newcommand*{\Mpm}{\partialmolar{M}}
\newcommand*{\Bpm}{\partialmolar{B}}
```

\cPpm \cVpm

Partial molar heat capacities are *hard,* but the following implementation seems to work flawlessly. . . so far.

```
\newcommand*{\cPpm}{%
  \def\@@@generic@pm##1##2{%
    \thermo@overline{\pm@symbol^{##2}_{\pressure@symbol,\pm@arg}}}%
  \def\@@@@generic@pm{\thermo@overline{\pm@symbol_{\pressure@symbol,\pm@arg}}}%
  \def\@@@@@generic@pm##1##2##3##4{%
    \thermo@overline{\pm@symbol^{##2}_{\pressure@symbol,##4}}}%
  \def\@pm@bracket@yes[##1]##2{%
    \thermo@overline{\pm@symbol^{##1}_{\pressure@symbol,##2}}}%
  \def\@pm@bracket@no##1{\thermo@overline{\pm@symbol_{\pressure@symbol,##1}}}%
  \partialmolar{\heatcapacitysymbol}%
}
\newcommand*{\cVpm}{%
  \def\@@@generic@pm##1##2{%
    \thermo@overline{\pm@symbol^{##2}_{\volume@symbol,\pm@arg}}}%
  \def\@@@@generic@pm{\thermo@overline{\pm@symbol_{\volume@symbol,\pm@arg}}}%
  \def\@@@@@generic@pm##1##2##3##4{%
    \thermo@overline{\pm@symbol^{##2}_{\volume@symbol,##4}}}%
  \def\@pm@bracket@yes[##1]##2{%
    \thermo@overline{\pm@symbol^{##1}_{\volume@symbol,##2}}}%
  \def\@pm@bracket@no##1{\thermo@overline{\pm@symbol_{\volume@symbol,##1}}}%
```

```
  \partialmolar{\heatcapacitysymbol}%
}
```

4.9 Residual and Excess Properties

\residual
\excess

Macros are defined for residual properties (departure from non-ideal gases) and excess properties (departure from ideal solutions). We begin with two macros to use for defining generic residual and excess properties that are not already defined.

```
\newcommand*{\residual}{R}
\newcommand*{\excess}{E}
```

\UR
\HR
\FR
\GR
\VR
\SR

The ordinary residual properties are molar.

```
\newcommand*{\UR}{\Um^\residual}
\newcommand*{\HR}{\Hm^\residual}
\newcommand*{\FR}{\Fm^\residual}
\newcommand*{\GR}{\Gm^\residual}
\newcommand*{\VR}{\Vm^\residual}
\newcommand*{\SR}{\Sm^\residual}
```

\URt
\HRt
\FRt
\GRt
\VRt
\SRt

We define similar macros for the extensive residual properties. These are appended by a `t` (for "total").

```
\newcommand*{\URt}{\Ut^\residual}
\newcommand*{\HRt}{\Ht^\residual}
\newcommand*{\FRt}{\Ft^\residual}
\newcommand*{\GRt}{\Gt^\residual}
\newcommand*{\VRt}{\Vt^\residual}
\newcommand*{\SRt}{\St^\residual}
```

\URs
\HRs
\FRs
\GRs
\VRs
\SRs

Similar macros are declared for the specific residual properties. These are appended by an `s`.

```
\newcommand*{\URs}{\Us^\residual}
\newcommand*{\HRs}{\Hs^\residual}
\newcommand*{\FRs}{\Fs^\residual}
\newcommand*{\GRs}{\Gs^\residual}
\newcommand*{\VRs}{\Vs^\residual}
\newcommand*{\SRs}{\Ss^\residual}
```

\URpm
\HRpm
\FRpm
\GRpm
\VRpm
\SRpm

Now for the partial molar residual properties. These are appended by `pm`.

```
\newcommand*{\URpm}{\partialmolar{\internalenergy@symbol^\residual}}
\newcommand*{\HRpm}{\partialmolar{\enthalpy@symbol^\residual}}
\newcommand*{\FRpm}{\partialmolar{\Helmholtz@symbol^\residual}}
\newcommand*{\GRpm}{\partialmolar{\Gibbs@symbol^\residual}}
\newcommand*{\VRpm}{\partialmolar{\volume@symbol^\residual}}
\newcommand*{\SRpm}{\partialmolar{\entropy@symbol^\residual}}
```

\UE
\HE
\FE
\GE
\VE
\SE

Now for the excess molar properties.

```
\newcommand*{\UE}{\Um^\excess}
\newcommand*{\FE}{\Fm^\excess}
\newcommand*{\HE}{\Hm^\excess}
\newcommand*{\GE}{\Gm^\excess}
\newcommand*{\SE}{\Sm^\excess}
\newcommand*{\VE}{\Vm^\excess}
```

\UEt \HEt \FEt \GEt \VEt \SEt

Similarly, the extensive excess properties.

```
\newcommand*{\UEt}{\Ut^\excess}
\newcommand*{\FEt}{\Ft^\excess}
\newcommand*{\HEt}{\Ht^\excess}
\newcommand*{\GEt}{\Gt^\excess}
\newcommand*{\SEt}{\St^\excess}
\newcommand*{\VEt}{\Vt^\excess}
```

\UEs \HEs \FEs \GEs \VEs \SEs

Now for the specific excess properties.

```
\newcommand*{\UEs}{\Us^\excess}
\newcommand*{\FEs}{\Fs^\excess}
\newcommand*{\HEs}{\Hs^\excess}
\newcommand*{\GEs}{\Gs^\excess}
\newcommand*{\SEs}{\Ss^\excess}
\newcommand*{\VEs}{\Vs^\excess}
```

\UEpm \HEpm \FEpm \GEpm \VEpm \SEpm

Finally, the excess partial molar quantities.

```
\newcommand*{\UEpm}{\partialmolar{\internalenergy@symbol^\excess}}
\newcommand*{\HEpm}{\partialmolar{\enthalpy@symbol^\excess}}
\newcommand*{\FEpm}{\partialmolar{\Helmholtz@symbol^\excess}}
\newcommand*{\GEpm}{\partialmolar{\Gibbs@symbol^\excess}}
\newcommand*{\VEpm}{\partialmolar{\volume@symbol^\excess}}
\newcommand*{\SEpm}{\partialmolar{\entropy@symbol^\excess}}
```

\prodall \sumall \sumallbutlast

The `\sumall` command and its cousin, `\sumallbutlast`, simplify the typesetting of commonly-used sums; the command `\prodall` does the same thing for products, viz.,

```
\[ \sumall_i x_i = 1 \qquad \sumallbutlast_i x_i = 1 - x_\ncomponents
    \qquad K = \exp\left(\frac{-\Deltarxn\Gm^\std}{RT}\right)
             = \prodall_i a_i^{\nu_i} \]
```

gives

$$\sum_{i=1}^{C} x_i = 1 \qquad \sum_{i=1}^{C-1} x_i = 1 - x_C \qquad K = \exp\left(\frac{-\Delta G^\circ_{\text{rxn}}}{RT}\right) = \prod_{i=1}^{C} a_i^{\nu_i}$$

```
\newcommand*{\sumall}{\@ifnextchar_\@sumall\@@sumall}
\def\@sumall#1#2{\sum_{#2=1}^{\ncomponents}}
\def\@@sumall#1{\sum_{#1=1}^{\ncomponents}}
\newcommand*{\sumallbutlast}{\@ifnextchar_\@sumallbutlast\@@sumallbutlast}
\def\@sumallbutlast#1#2{\sum_{#2=1}^{\ncomponents-1}}
\def\@@sumallbutlast#1{\sum_{#1=1}^{\ncomponents-1}}
\newcommand*{\prodall}{\@ifnextchar_\@prodall\@@prodall}
\def\@prodall#1#2{\prod_{#2=1}^{\ncomponents}}
\def\@@prodall#1{\prod_{#1=1}^{\ncomponents}}
```

\IG \IGM \IS

The `\IG`, `\IGM`, and `\IS` macros (meaning "ideal gas," "ideal gas mixture," and "ideal solution," respectively) should be used to make clean transitions between textbooks—some use "IM" rather than "IS" for example.

```
\newcommand*{\IG}{{\text{IG}}}
\newcommand*{\IGM}{{\text{IGM}}}
\newcommand*{\IS}{{\text{IS}}}
```

\Henryrat
\Henrymol

The Henry's Law constants for the rational basis ($y_i P = x_i h_i$) and the molal basis ($y_i P = C_i \mathcal{H}_i$) are given by the macros `\Henryrat` and `\Henrymol`, respectively. Using them this way consistently allows for easy switching back and forth.

```
\newcommand*{\Henryrat}{h}
\newcommand*{\Henrymol}{{\mathcal{H}}}
```

\gammarat
\gammamol

The ordinary activity coefficient is universally denoted γ, so I have not defined a special macro for that. However, the Henry's Law activity coefficients are far from universal, so I have defined macros to make their use consistent. The defaults render `\gammarat` as γ^* and `\gammamol` as $\gamma^{\square}$. `\gammamol` will use `\square` from packages if it is defined; if not, it "fakes it" with the definition below (based on the amsthm package).

```
\AtBeginDocument{%
  \providecommand*{\square}{%
    \text{\leavevmode
      \hbox to.77778em{%
      \hfil\vrule
      \vbox to.6em{\hrule width.6em\vfil\hrule}%
      \vrule\hfil}%
    }%
  }%
}
\newcommand*{\gammarat}{\gamma^\ast}
\newcommand*{\gammamol}{\gamma^\square}
```

Jacobian

The `\Jacobian` command typesets the Jacobian, viz.,

```
\[ \Jacobian{K}{L}{X}{Y} =
  \begin{vmatrix}
    \displaystyle \Partial{K}{X}{Y} & \displaystyle \Partial{K}{Y}{X}
  \\[2ex]
    \displaystyle \Partial{L}{X}{Y} & \displaystyle \Partial{L}{Y}{X}
  \end{vmatrix}
\]
```

produces

$$\left(\frac{\partial(K,L)}{\partial(X,Y)}\right) = \begin{vmatrix} \left(\frac{\partial K}{\partial X}\right)_Y & \left(\frac{\partial K}{\partial Y}\right)_X \\ \left(\frac{\partial L}{\partial X}\right)_Y & \left(\frac{\partial L}{\partial Y}\right)_X \end{vmatrix}$$

This macro is defined purely for convenience.

```
\newcommand*{\Jacobian}[4]{\frac{\partial{(#1,#2)}}{\partial{(#3,#4)}}}%
```

Change History

v1.00

General: Initial public release 1

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Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

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